Enhancing Core Diffracted Arrivals using Super-virtual Interferometry

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Overview

1. Problem: Weak Event P-diff and S-diff Records Have Low SNR
   - Motivation
   - Method
   - Results
   - Random Noise Test
2. Problem: Transportable Array Stations Deployed After an Event
   - Motivation
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3. Conclusions

Next Section

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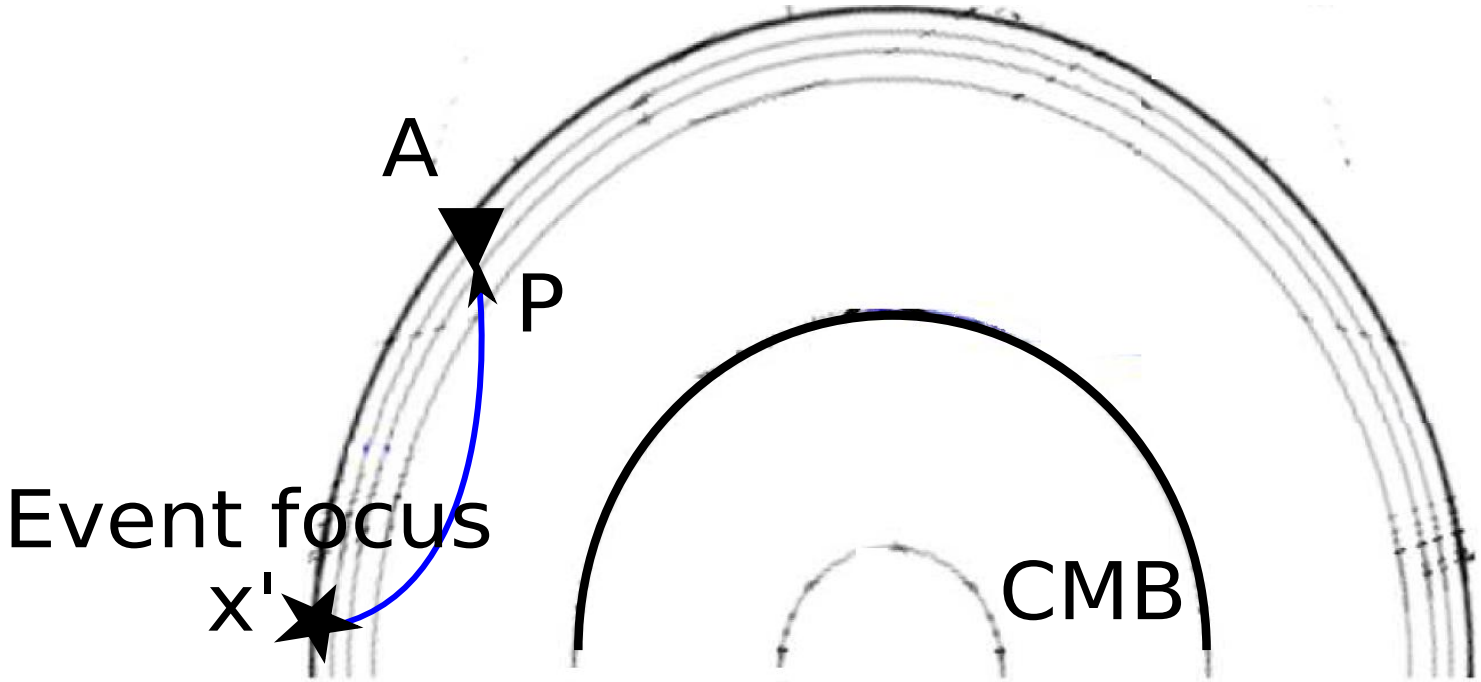
Core-Mantle Boundary Diffractions



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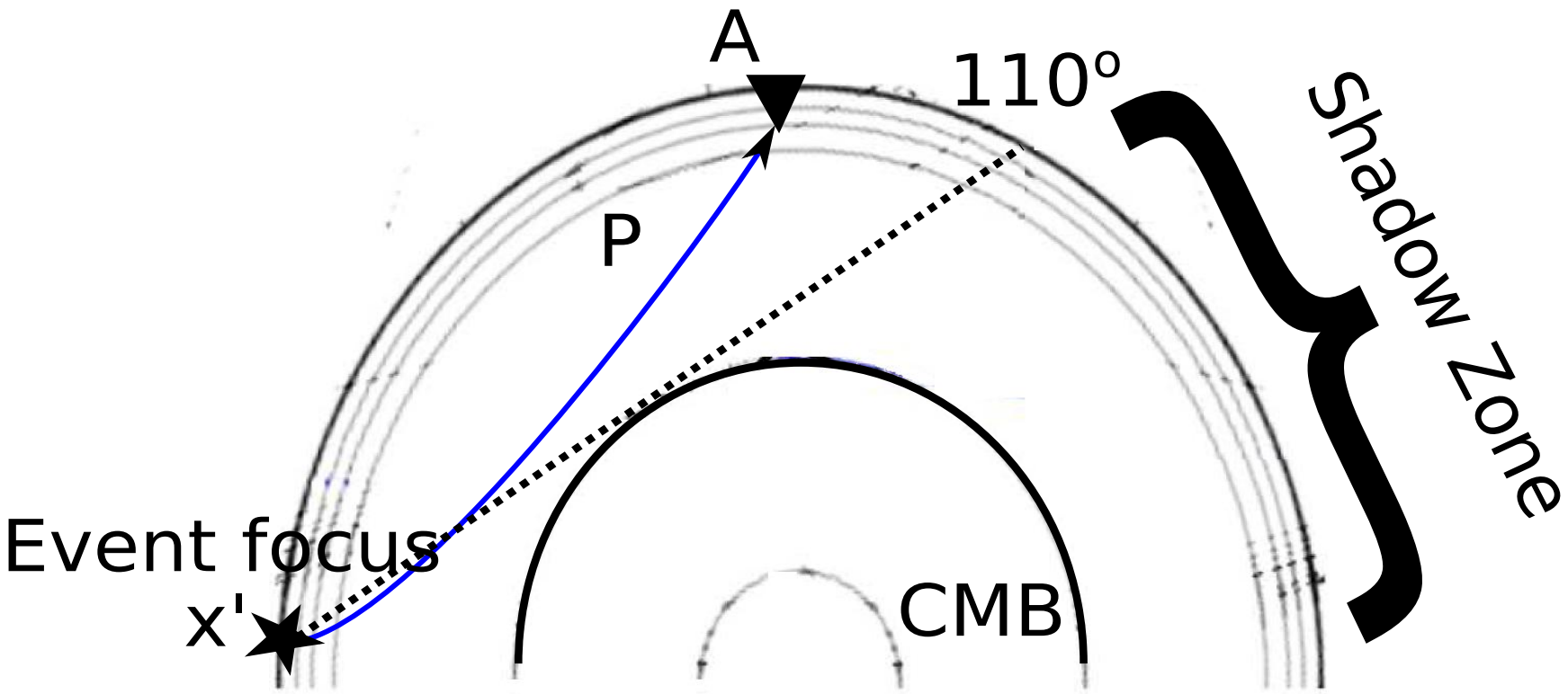
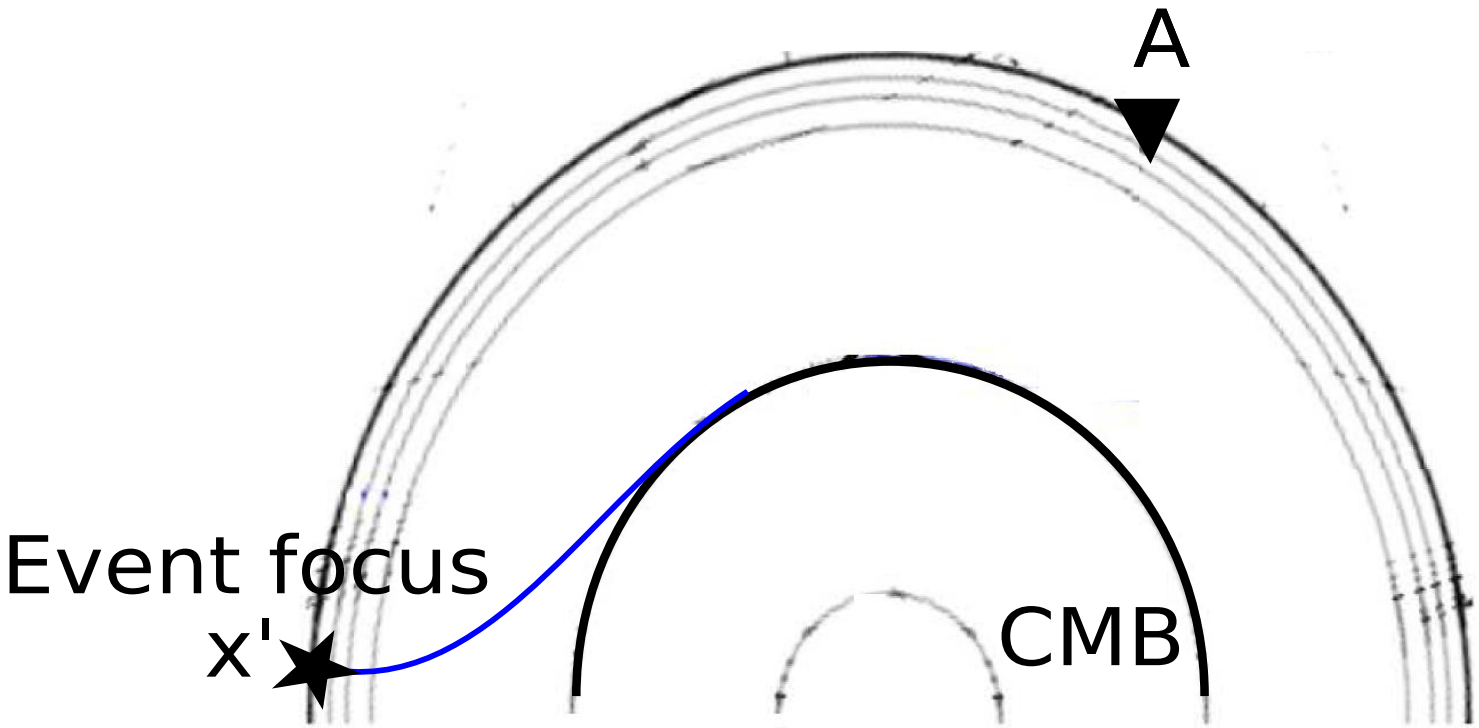


Pdiff and Sdiff Arrivals



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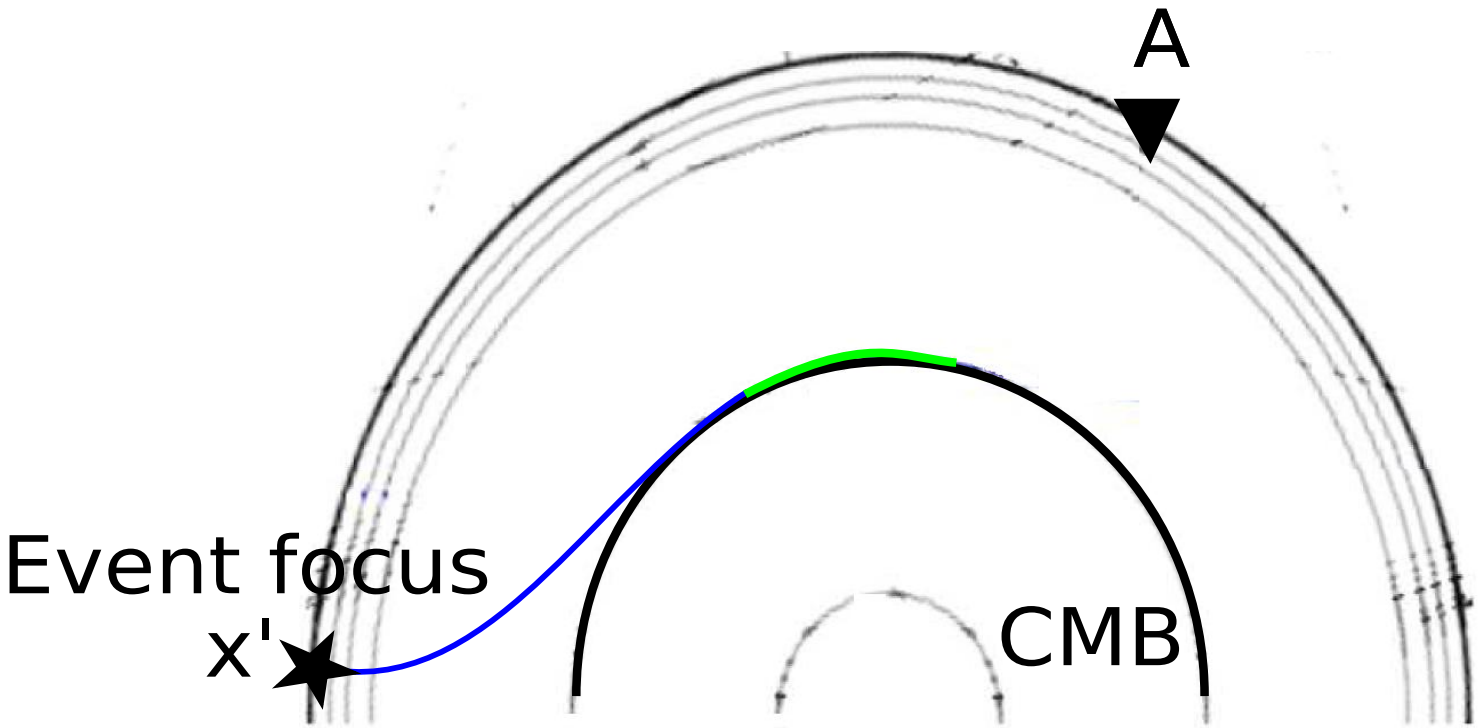

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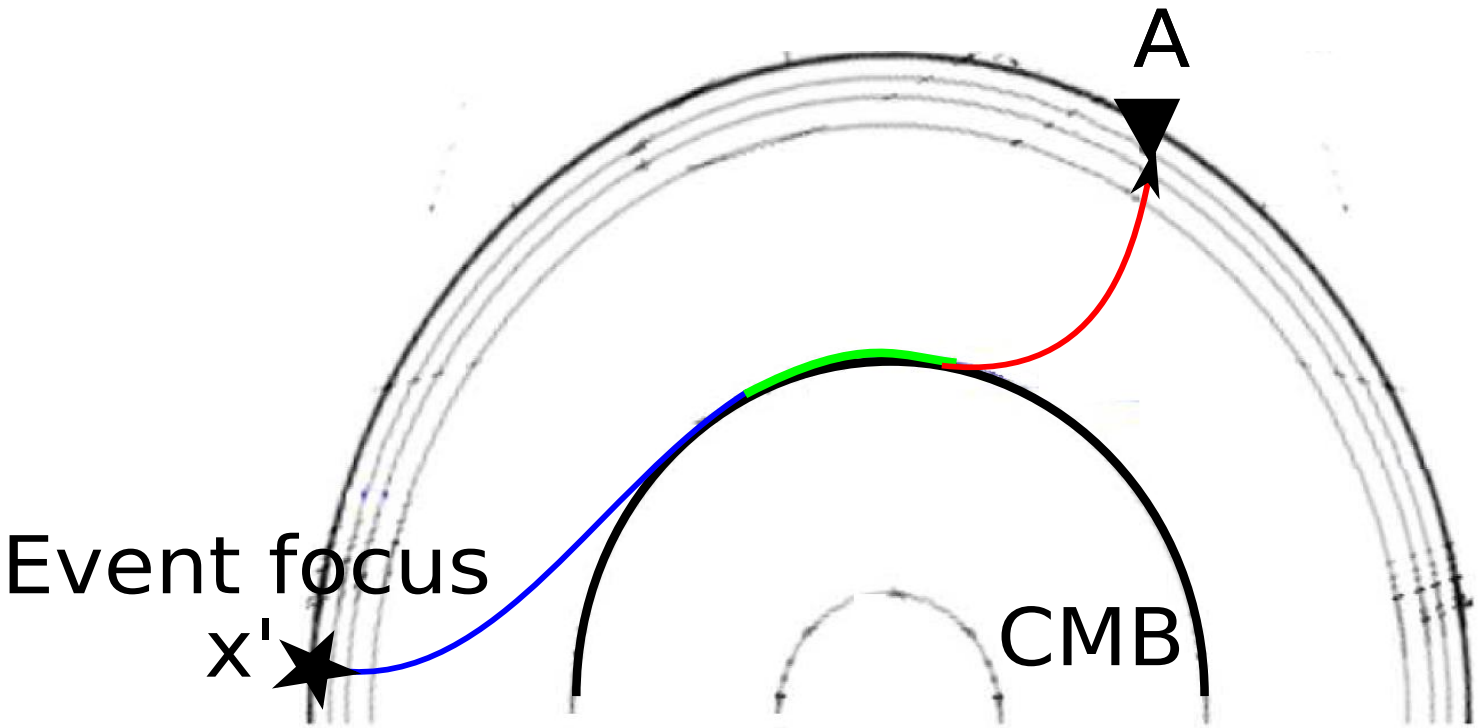
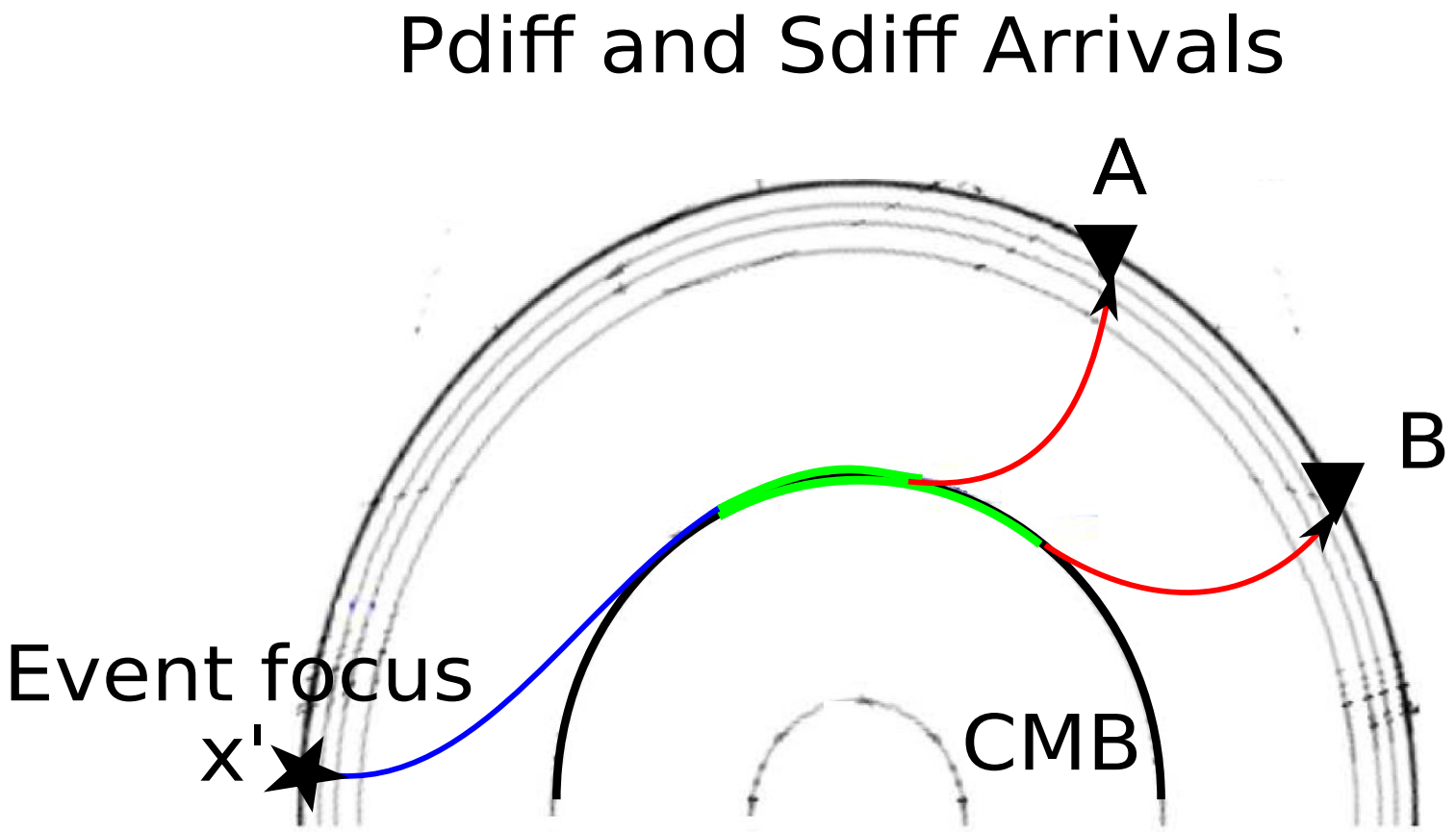
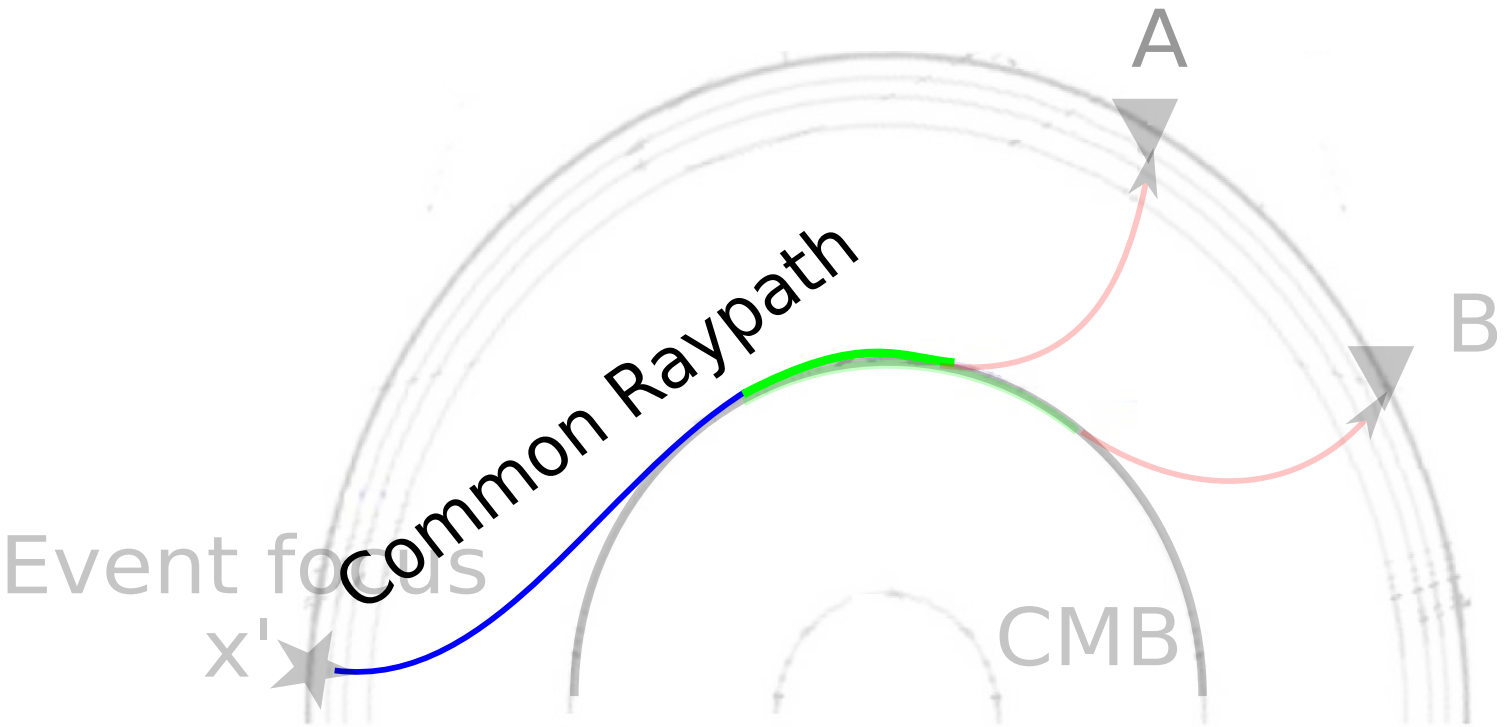
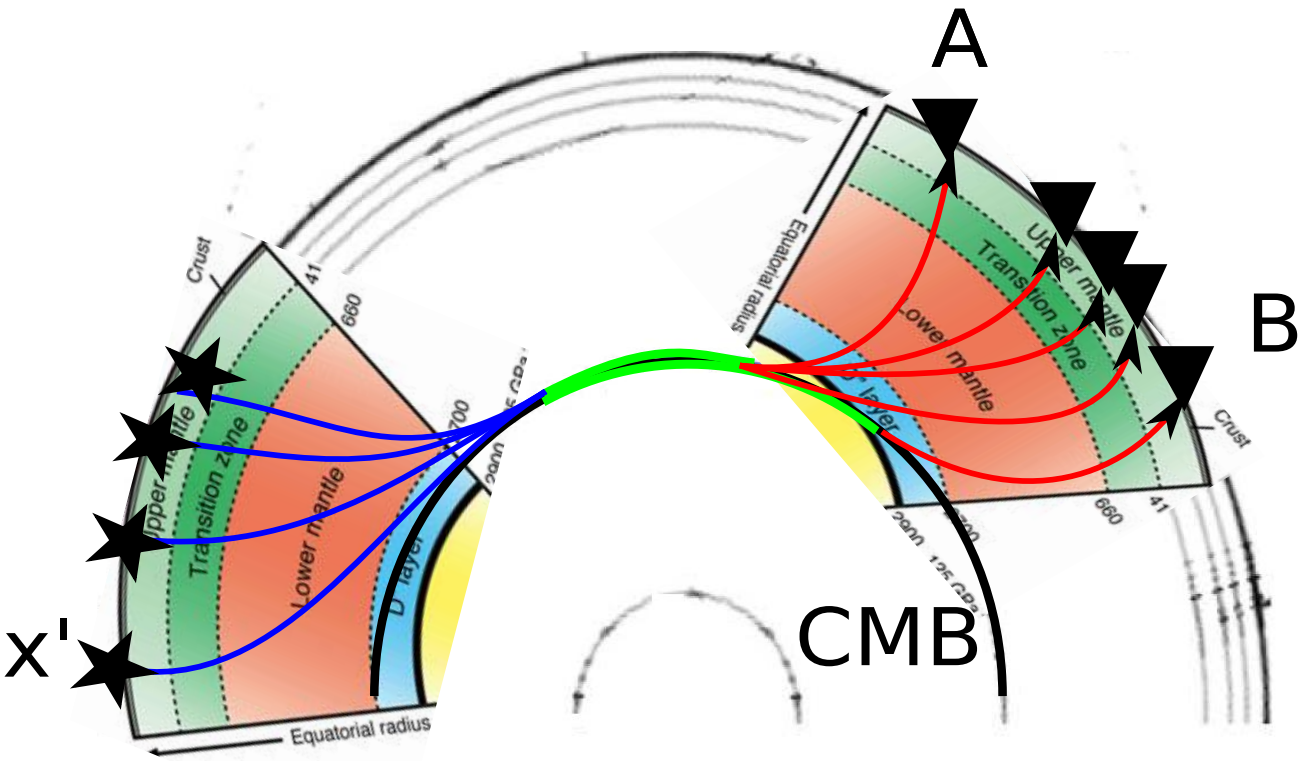



Pdiff and Sdiff Arrivals
A
B
Event focus
x'
CMB

Pdiff and Sdiff Arrivals



2D- Pdiff and Sdiff Tomography



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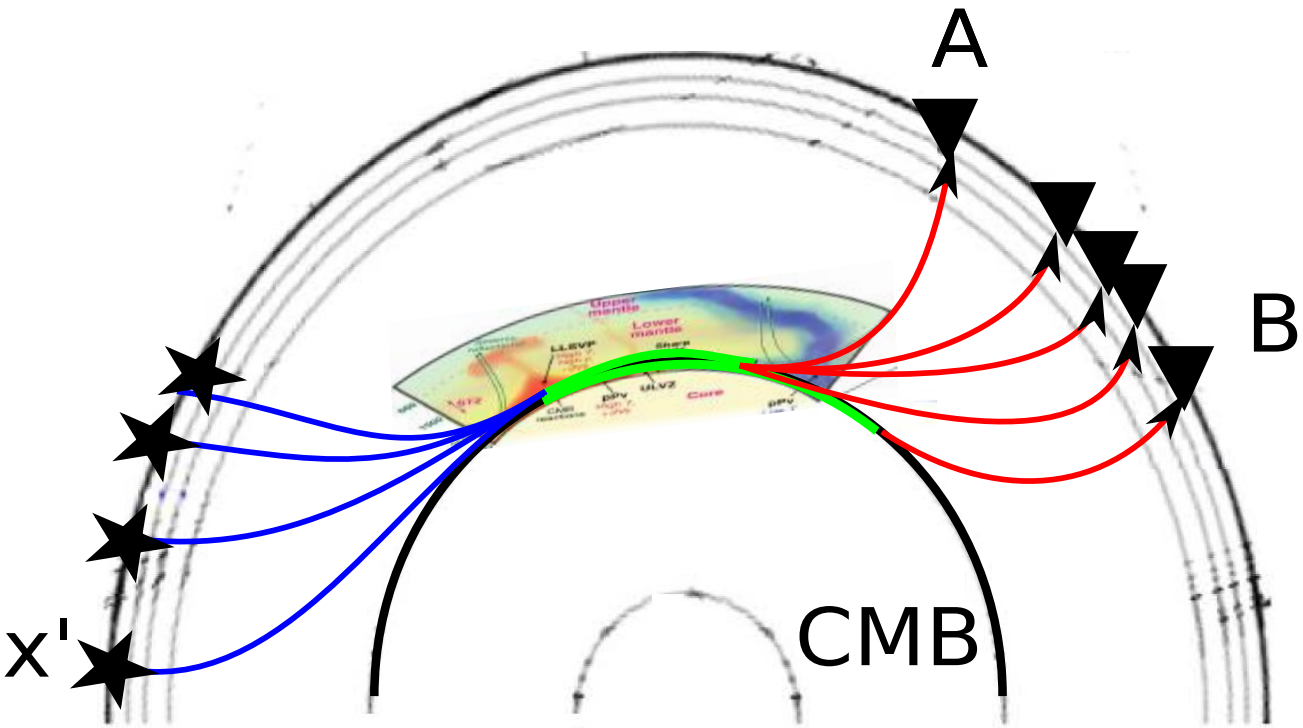

Tomography with P-diff and S-diff

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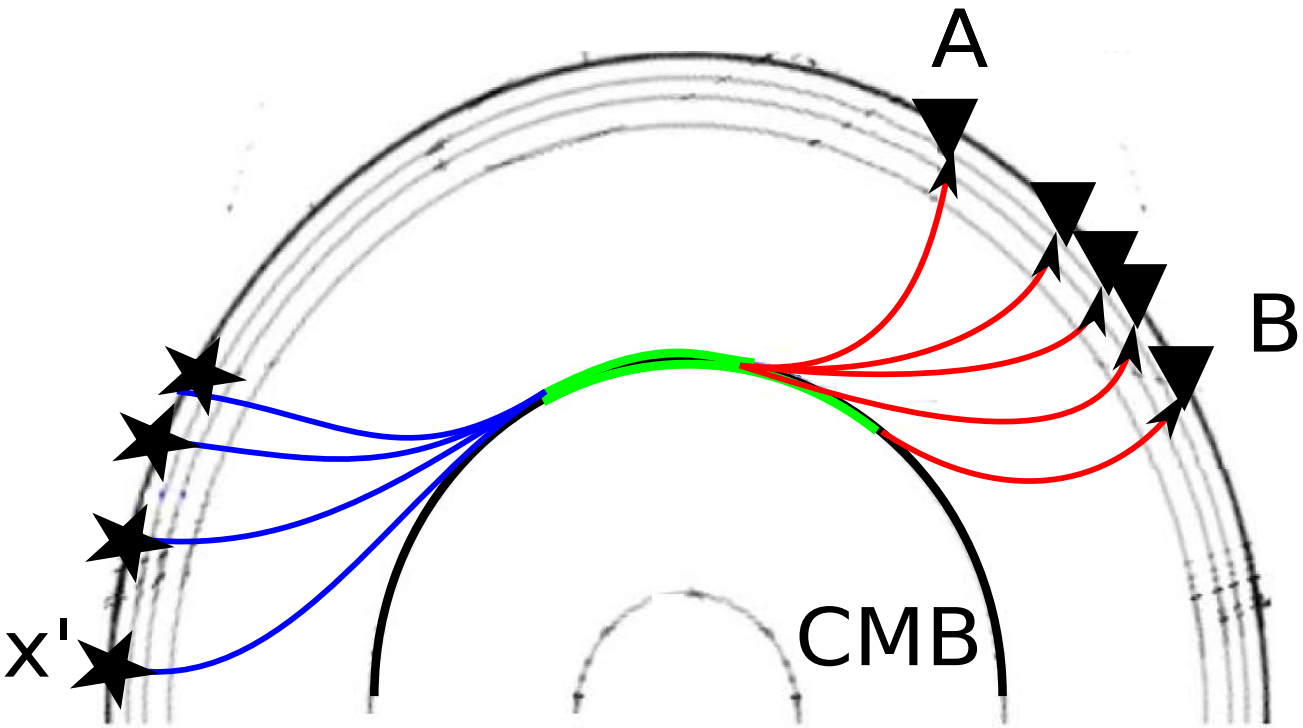
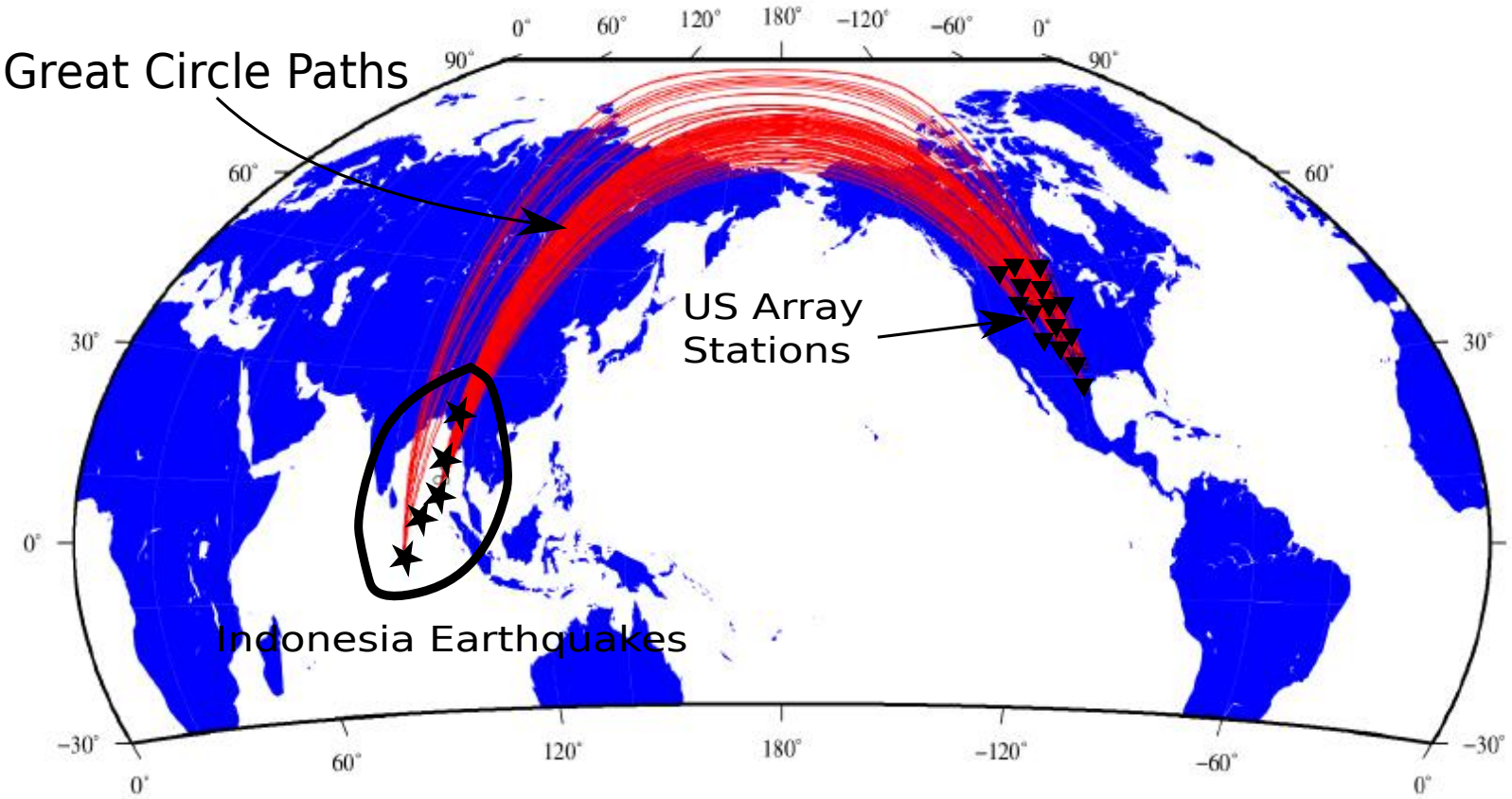
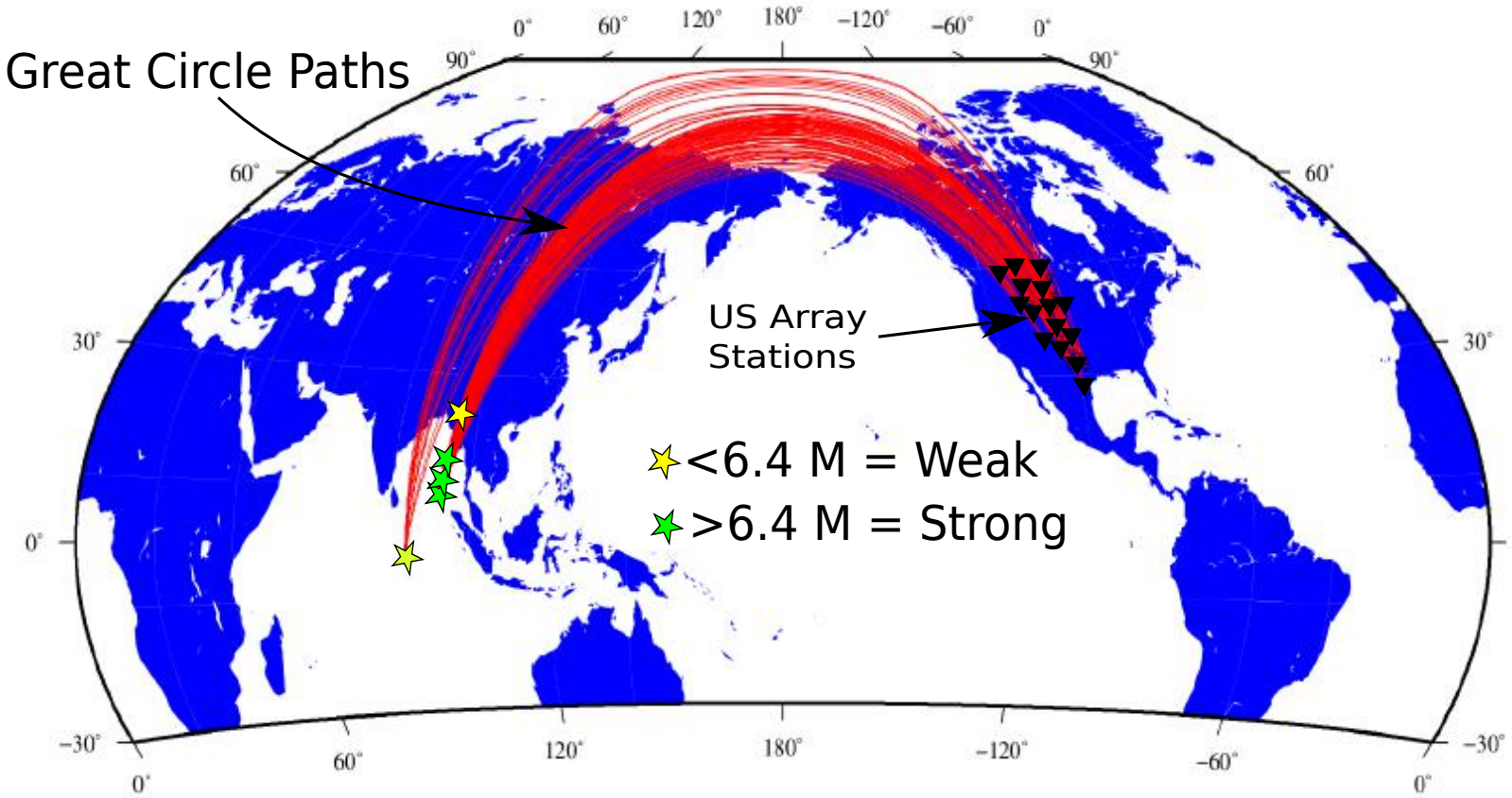


USArray: Many Receivers on a Same Great Circle



Points approx. on a great circle lie on same 2D plane

Very Few Localized Strong Magnitude Events



Events plotted between 2008 and 2010

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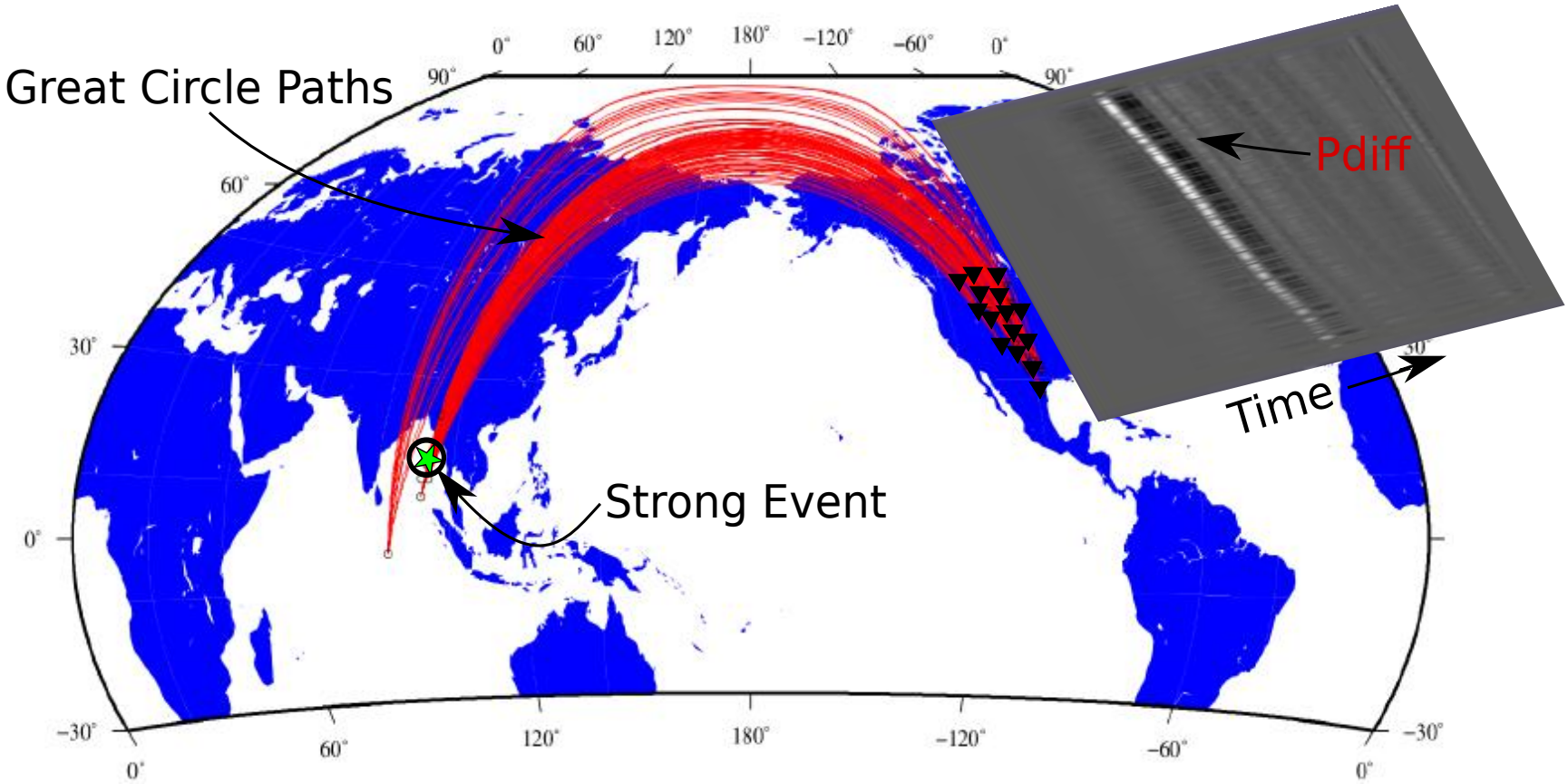

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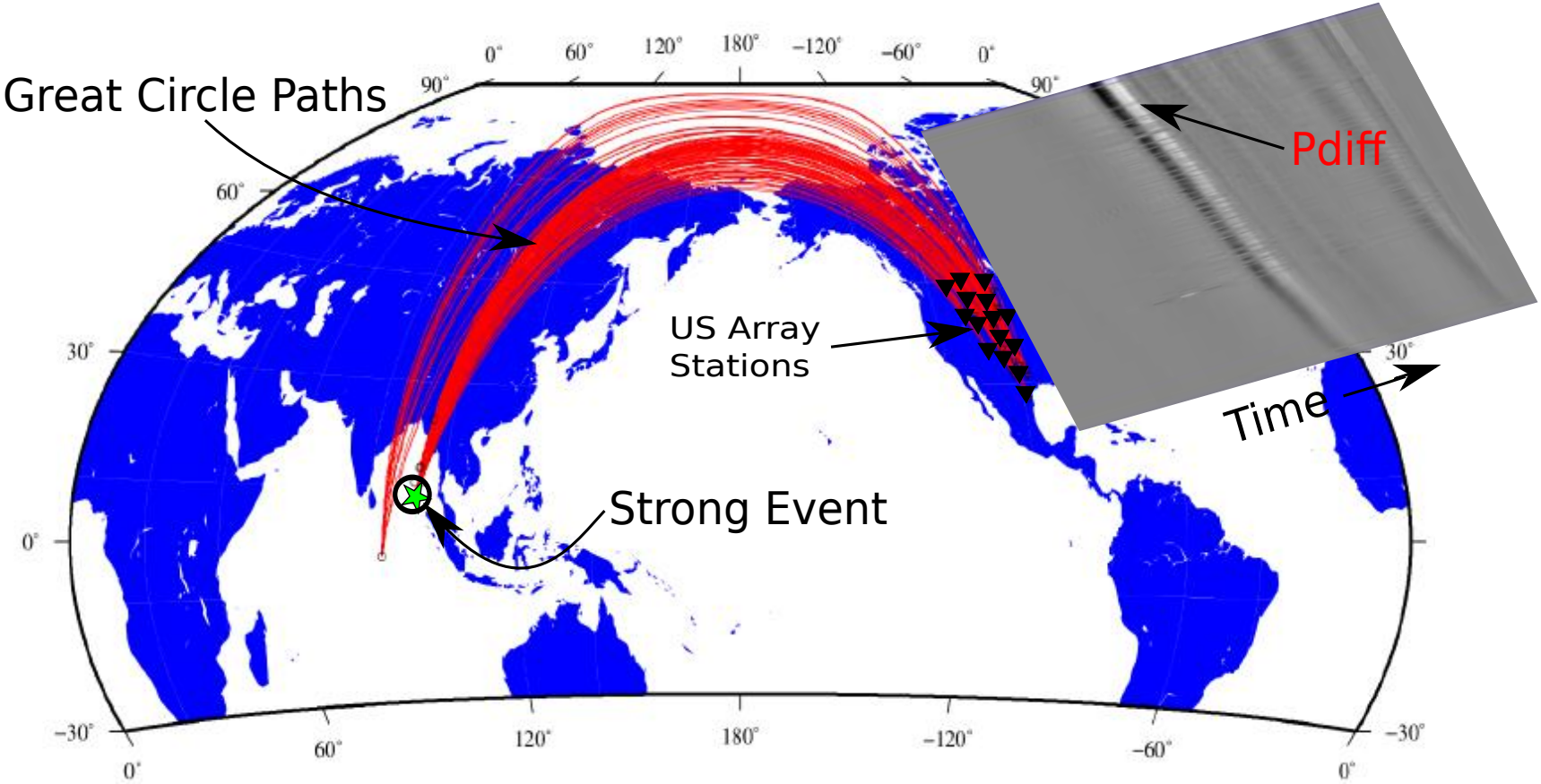

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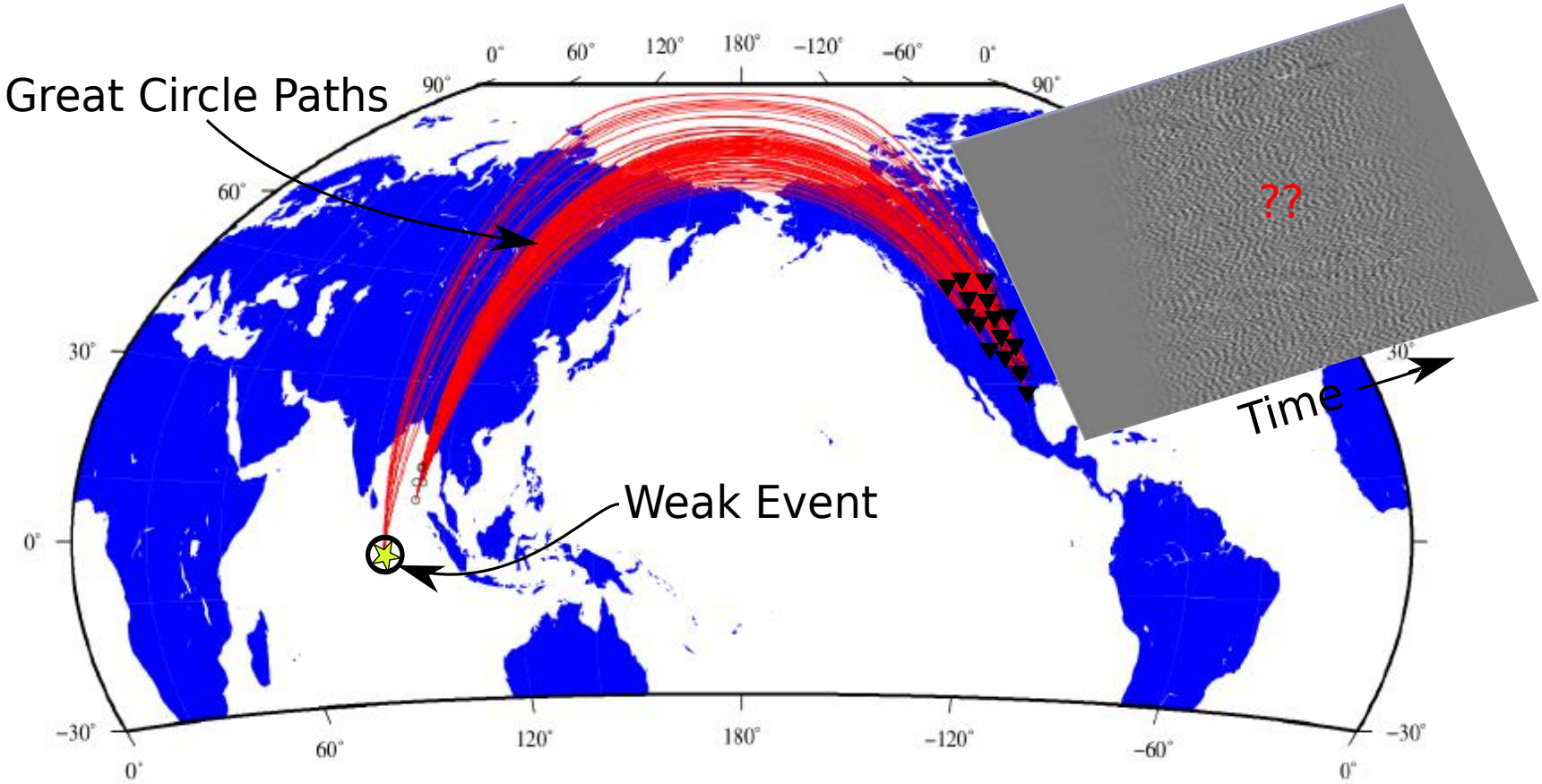

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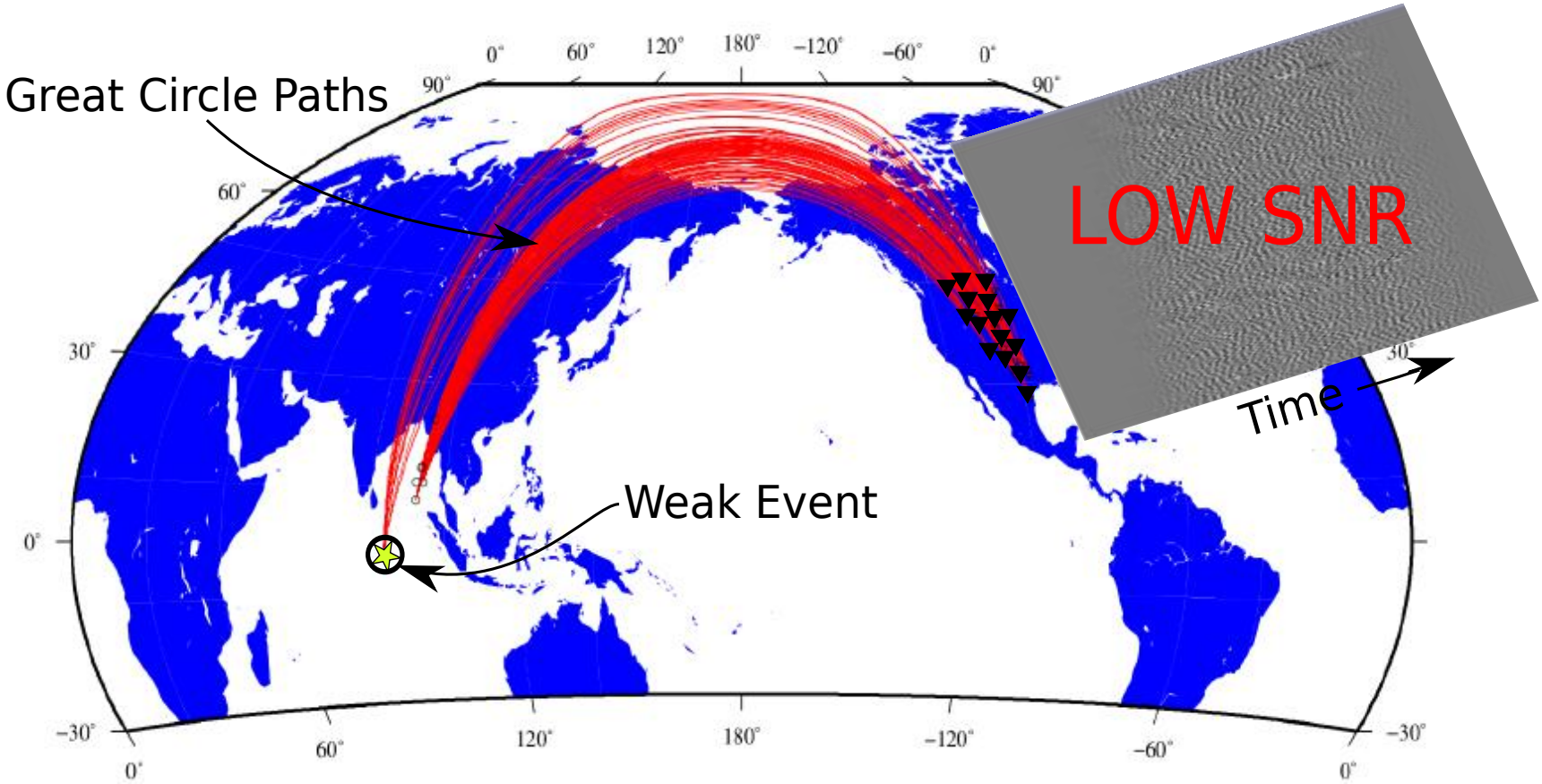

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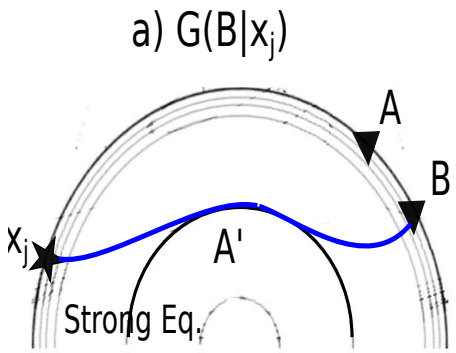

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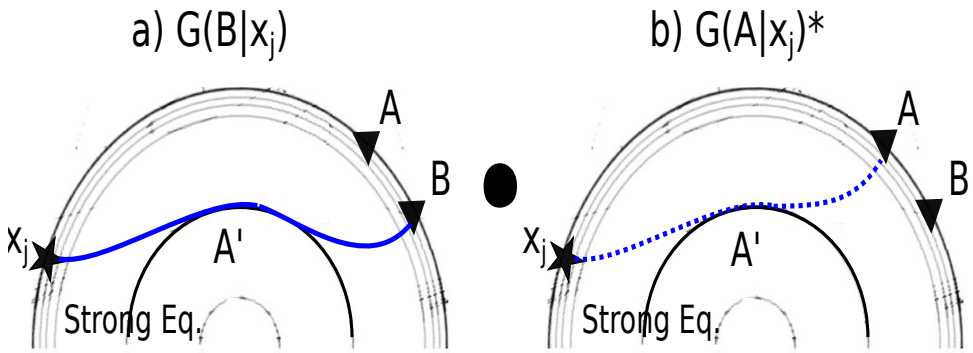

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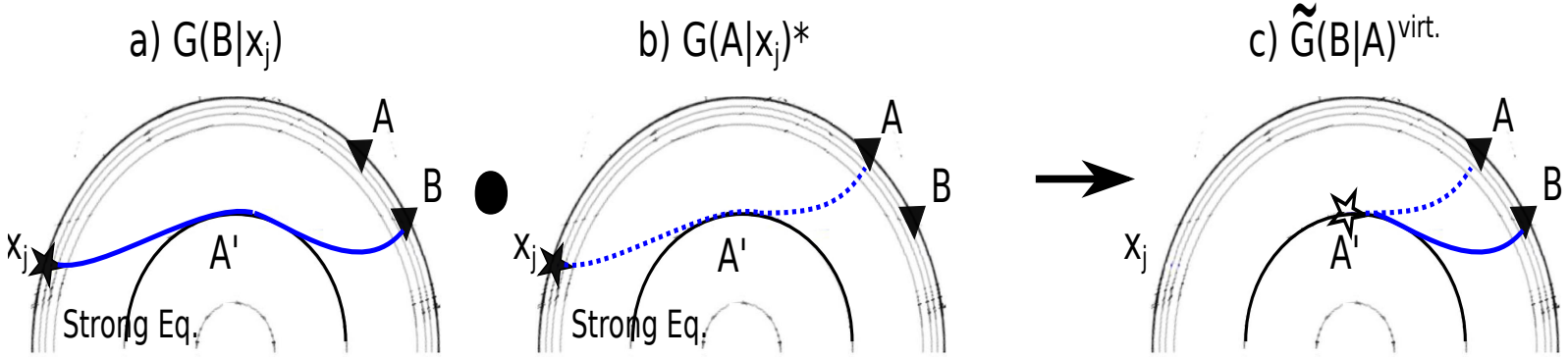

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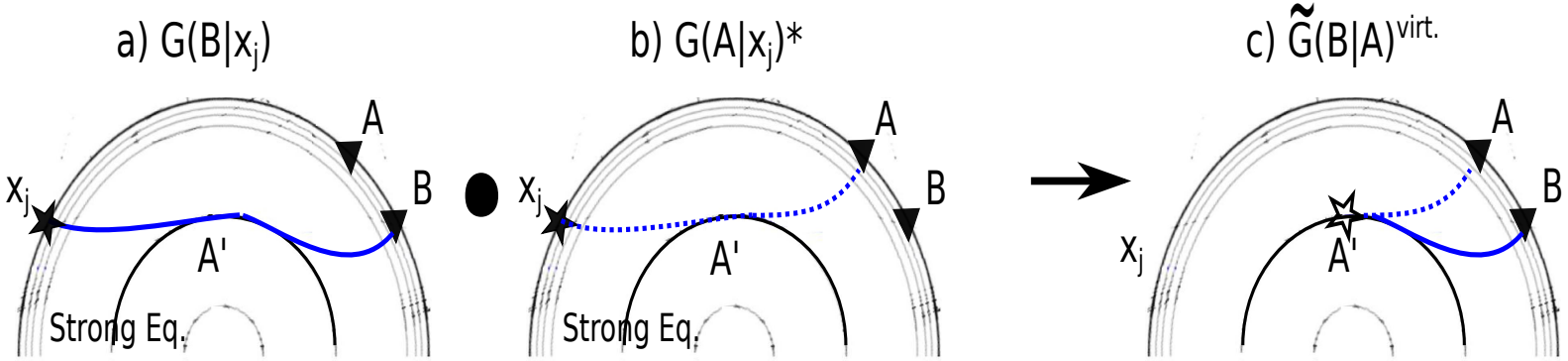

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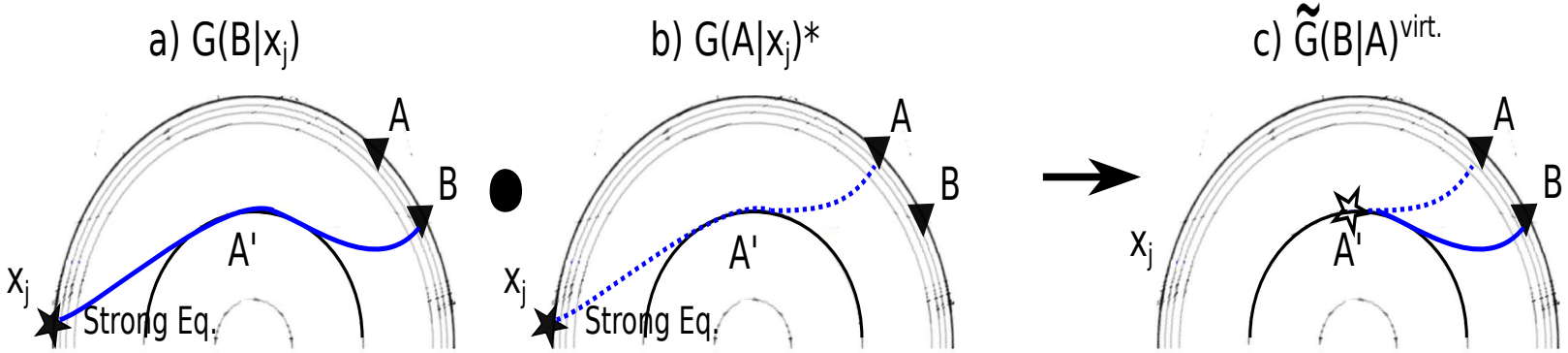

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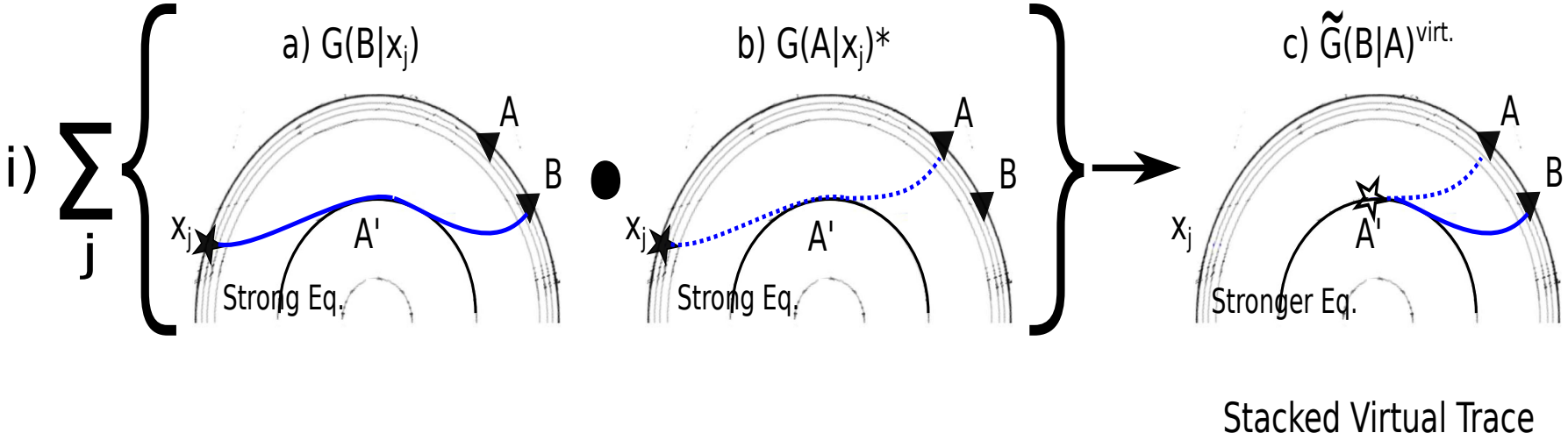

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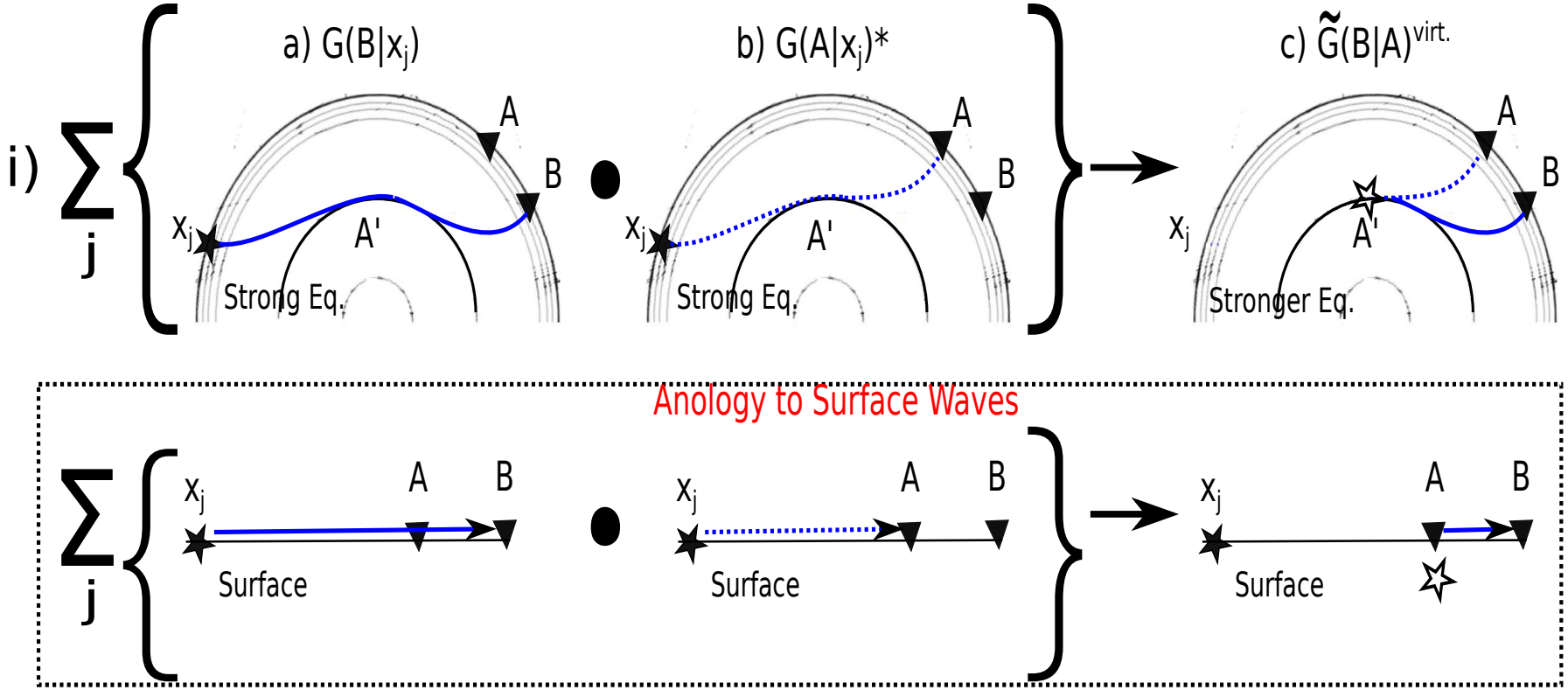

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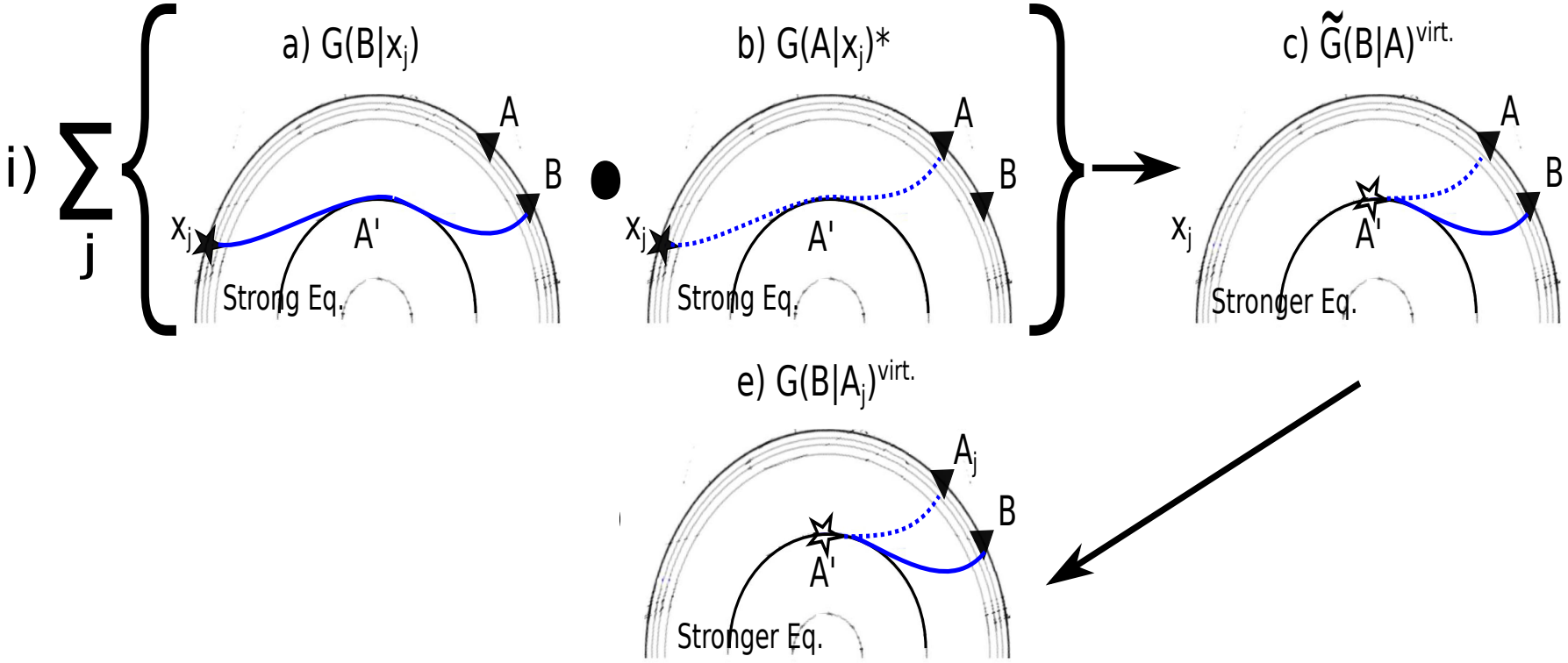

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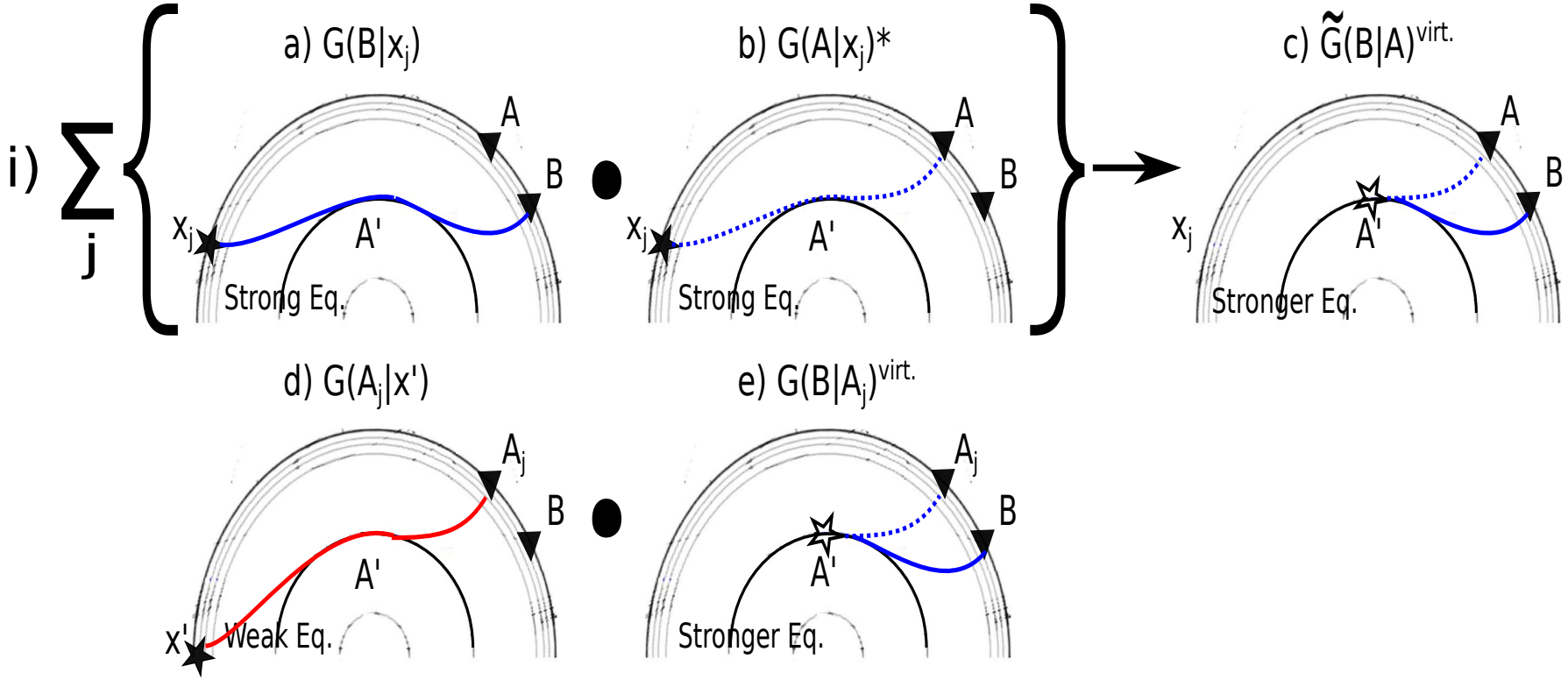

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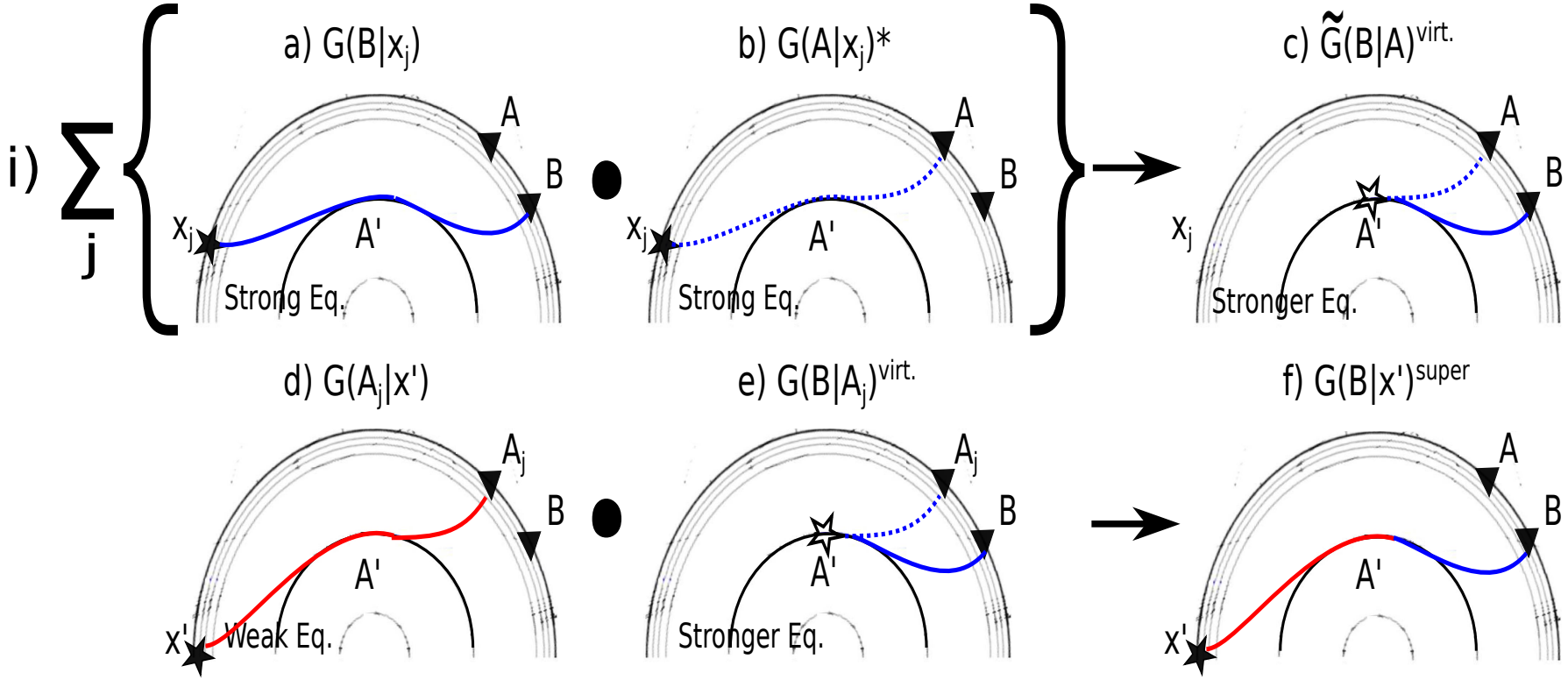

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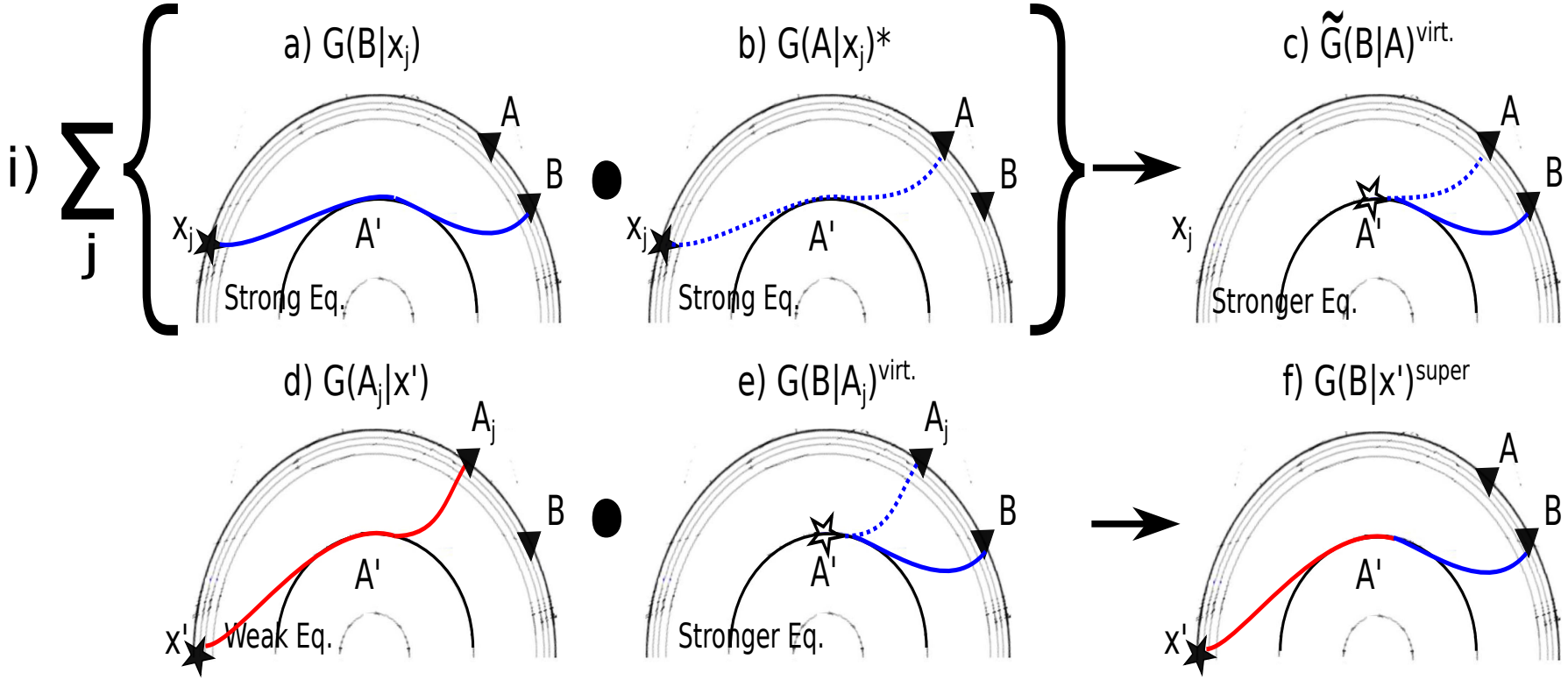

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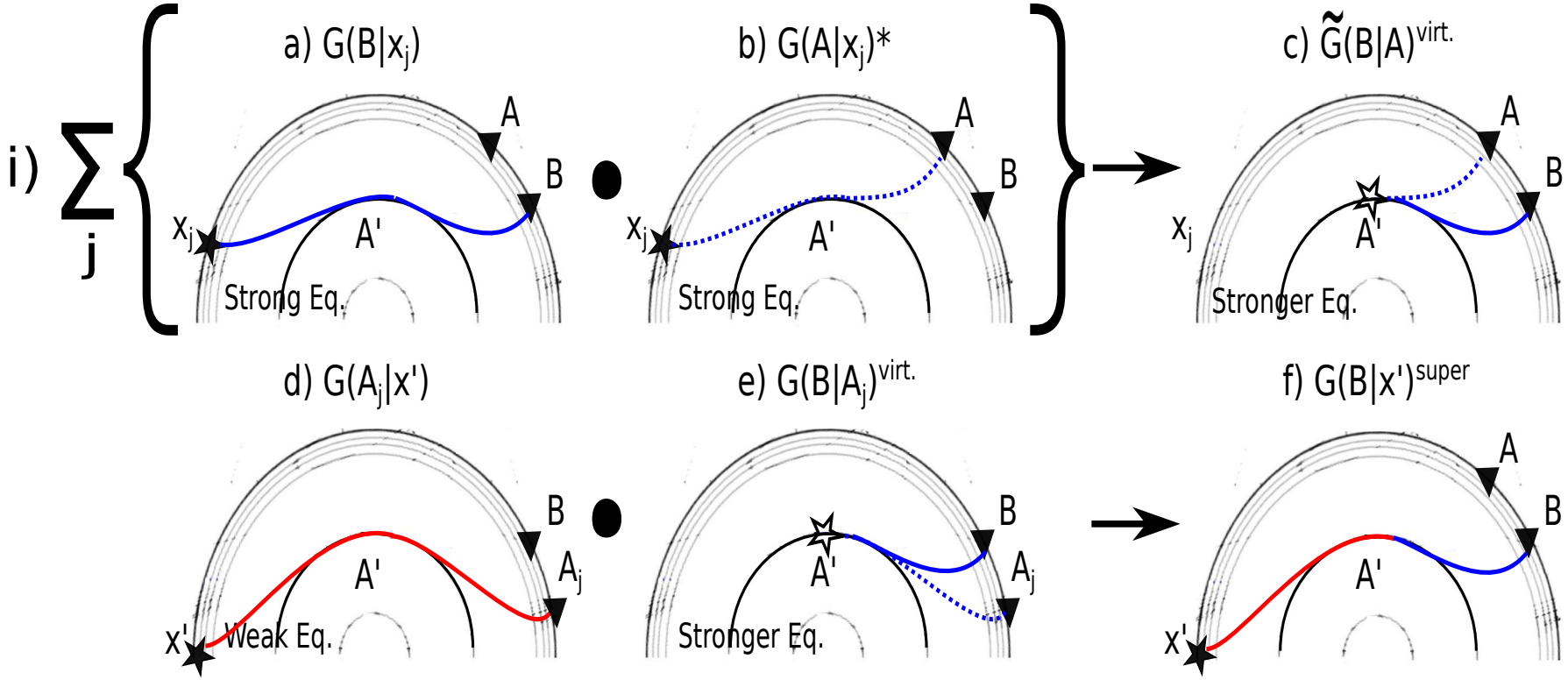

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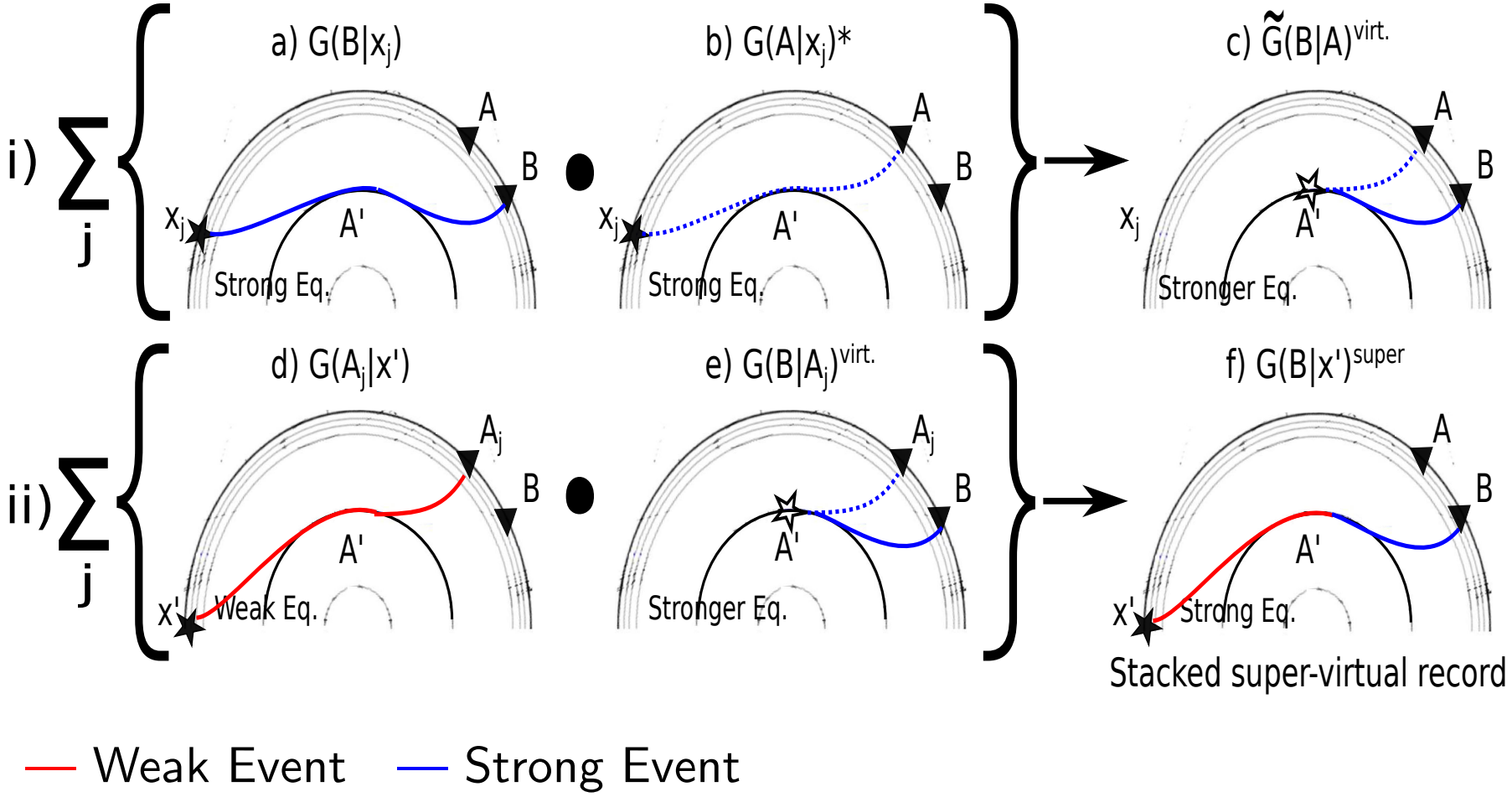

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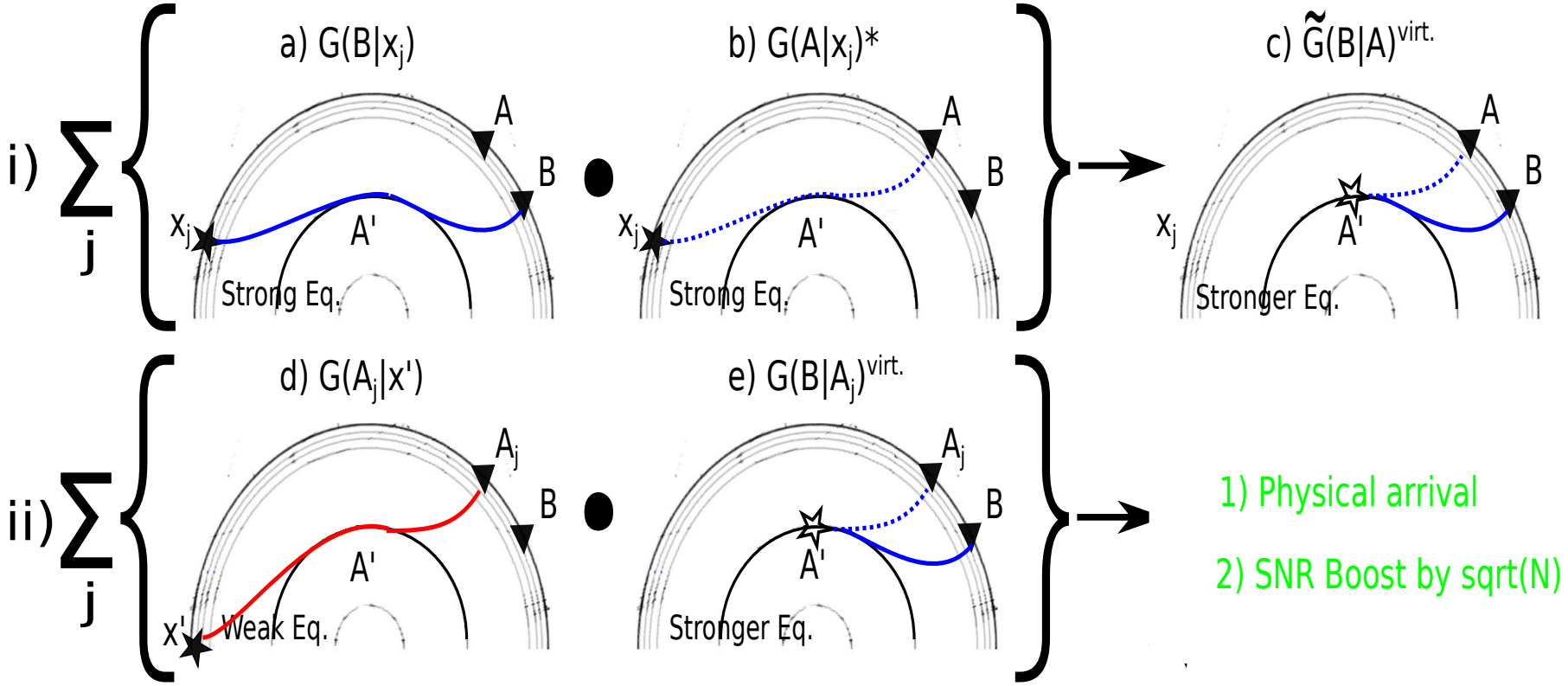

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Summary: Stacking Diffractions

1. How to increase SNR record at A ?
   - correct moveout to align N records
   - stack to boost SNR by $\sqrt{N}$.
2. How to get moveout information ?
   - [illegible]
   - [illegible]
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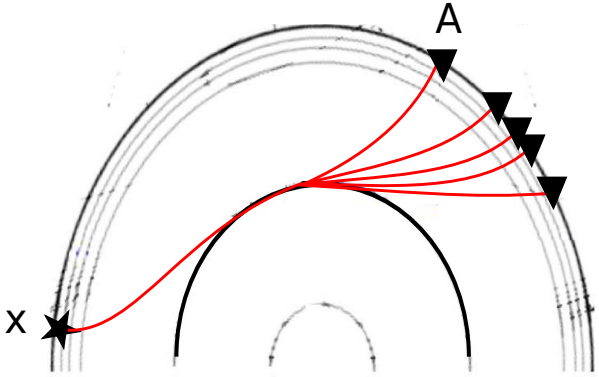
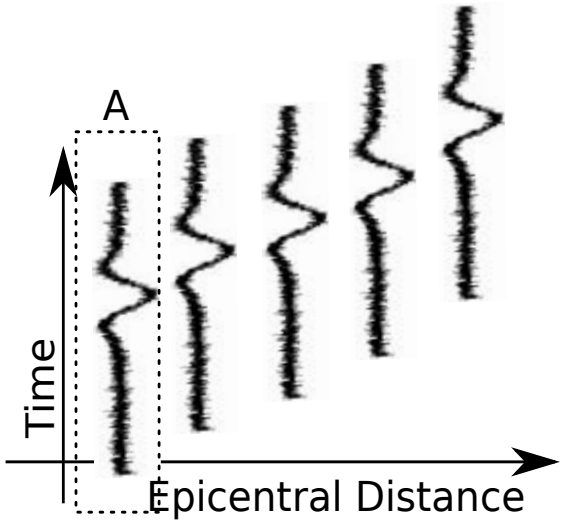
3. KEY IDEA: moveout of Pdiff (or) Sdiff arrivals in records independent of earthquake position





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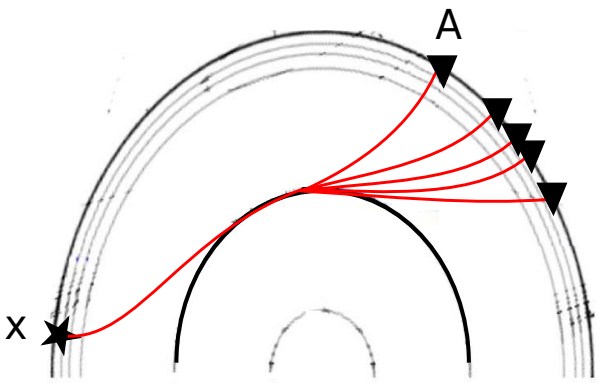
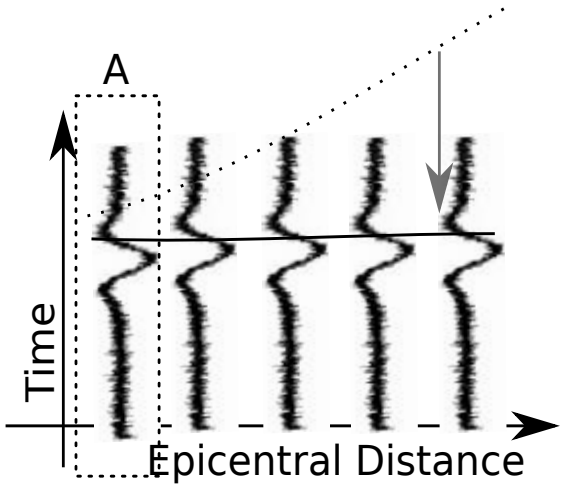
Summary: Stacking Diffractions

1. How to increase SNR record at A ?
   - correct moveout to align N records
   - stack to boost SNR by $\sqrt{N}$.
2. How to get moveout information ?
   - stack with different velocities
   - stack at different depths by cross-correlation
   - improve signal to noise ratio
   - measure moveout by cross-correlation
3. KEY IDEA: moveout of Pdiff (or) Sdiff arrivals in records independent of earthquake position





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   - stack to boost SNR by $\sqrt{N}$.
2. How to get moveout information ?
   - select a strong event
   - pick arrival times by cross-correlation
   - propagate to weak events
   - measure moveout curve
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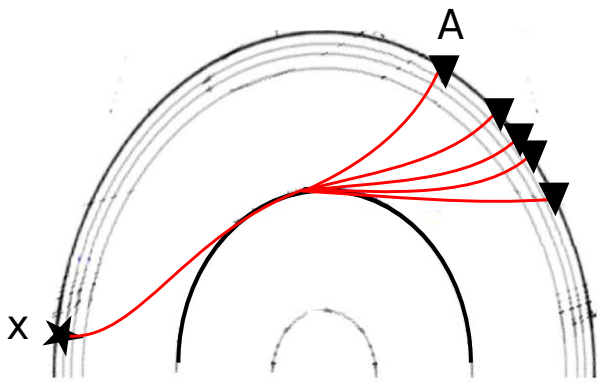

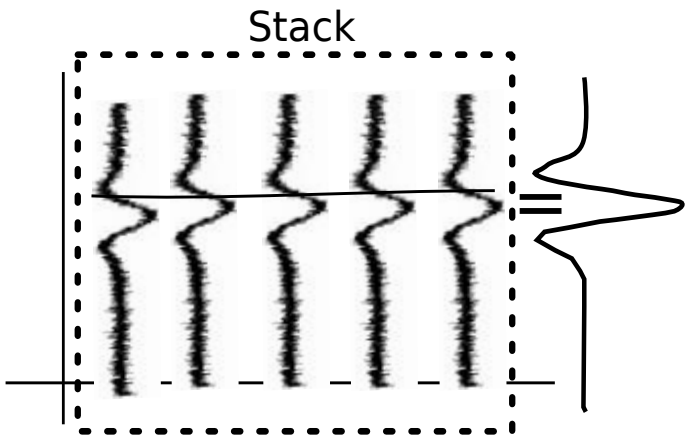

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   - moveout in weak Eq. records = moveout in strong Eq. records
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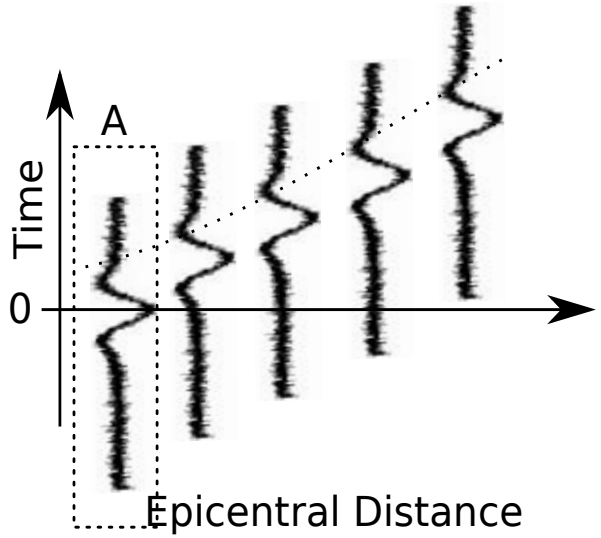

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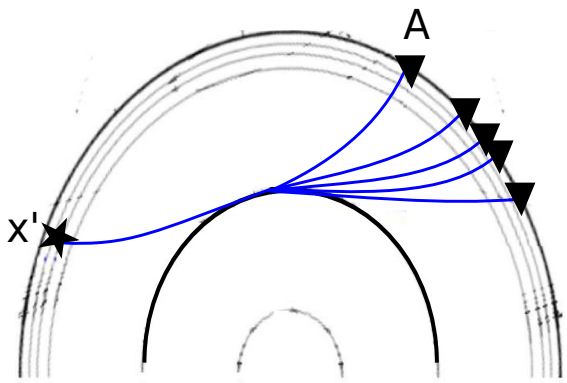

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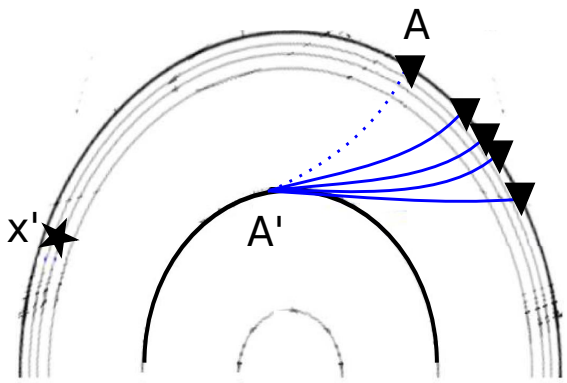

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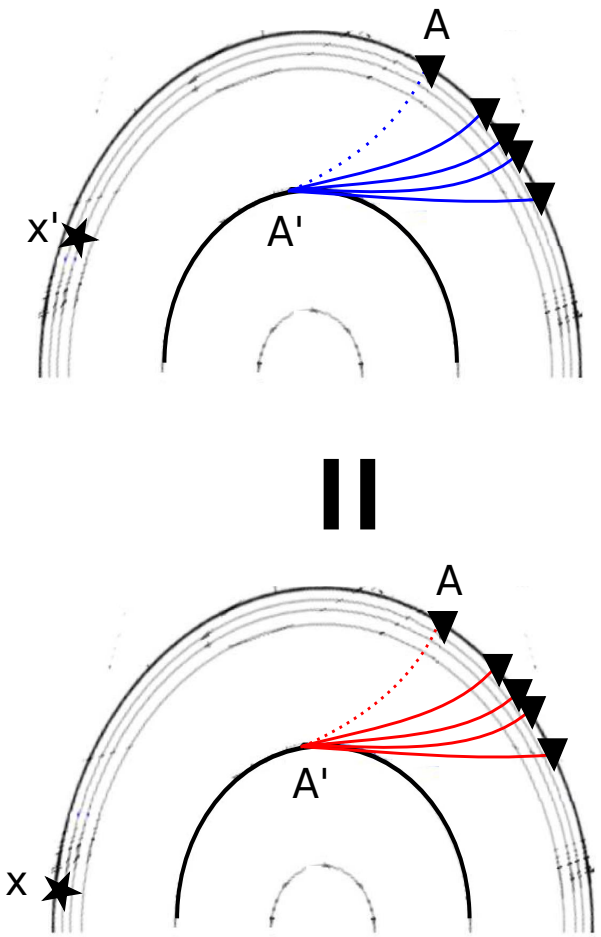

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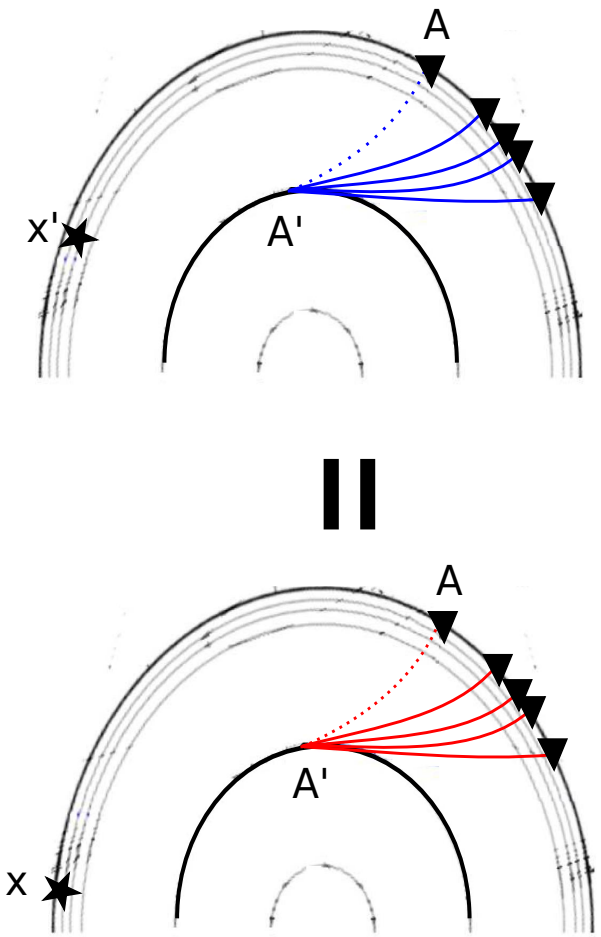

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Summary: Processing Workflow

1. Tapering, Demeaning, Re-sampling and Band pass filtering (0.008-0.12 Hz)
2. Deconvolution and a 400 sec time window
3. Virtual records by cross-correlation and stacking over sources.
4. Super-virtual records by convolution and stacking over stations (crucial step).
5. $\sqrt{N_g}$ enhancement of weak event records.

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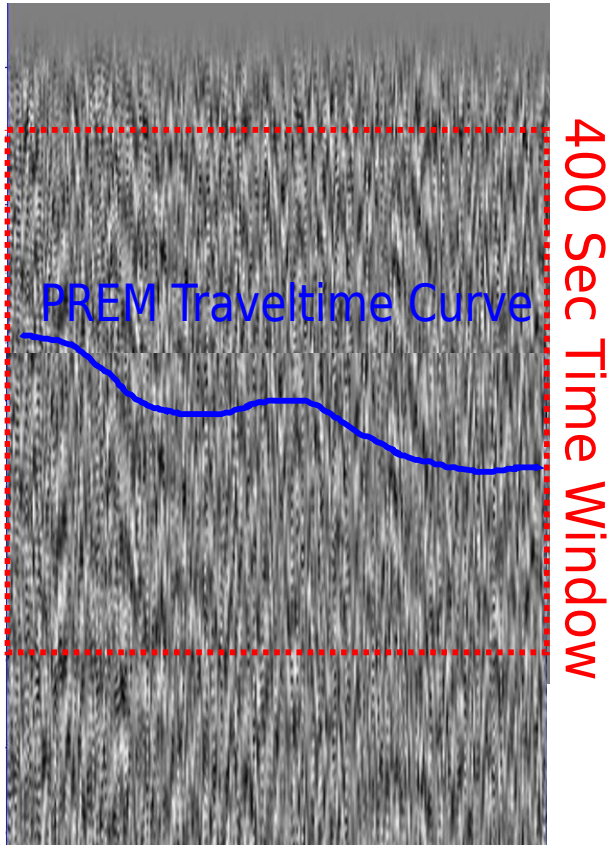

— Weak Event
— Strong Event

Summary: Processing Workflow

1. Tapering, Demeaning, Re-sampling and Band pass filtering (0.008-0.12 Hz)
2. Deconvolution and a 400 sec time window
3. Virtual records by cross-correlation and stacking over sources.
4. Super-virtual records by convolution and stacking over stations (crucial step).
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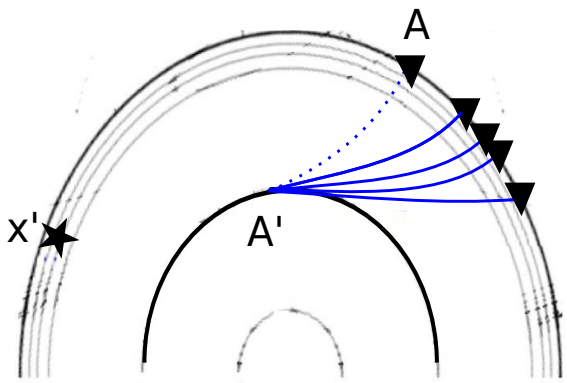

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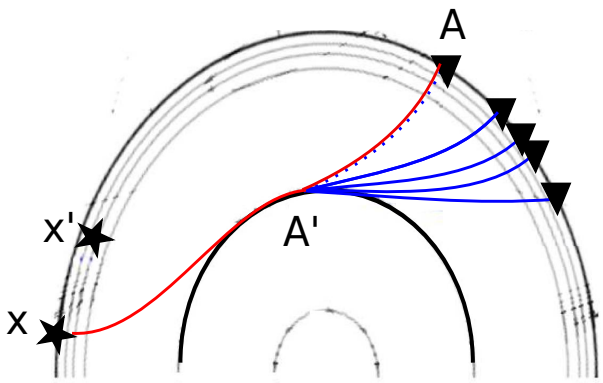

— Weak Event
— Strong Event

Summary: Processing Workflow

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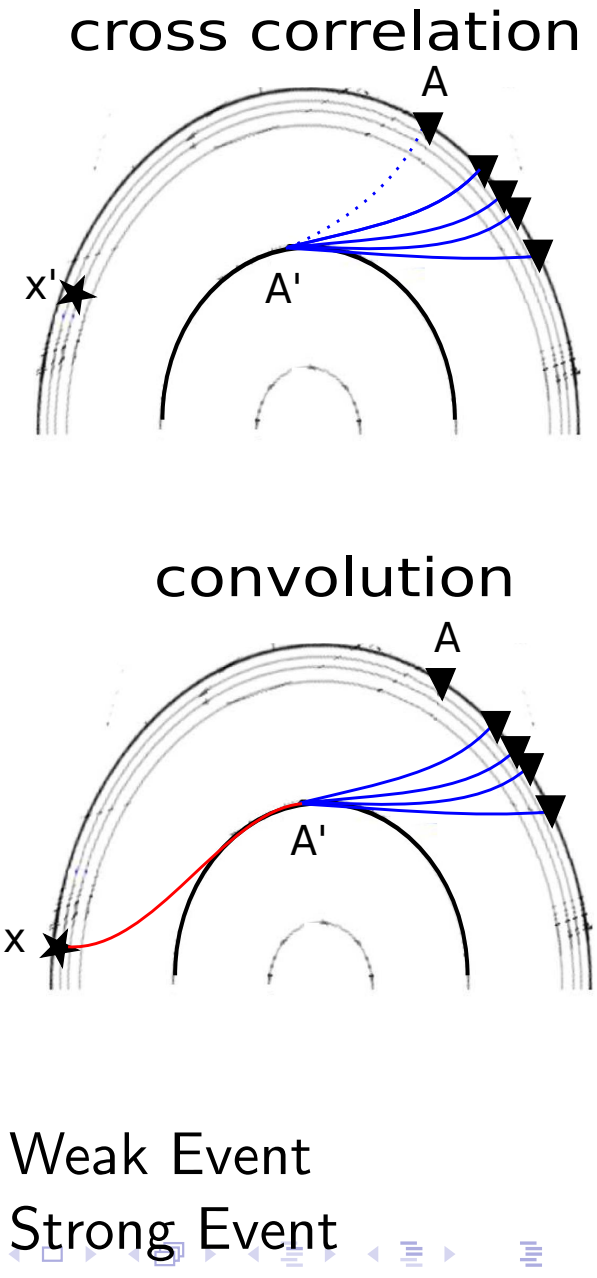

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   - Method
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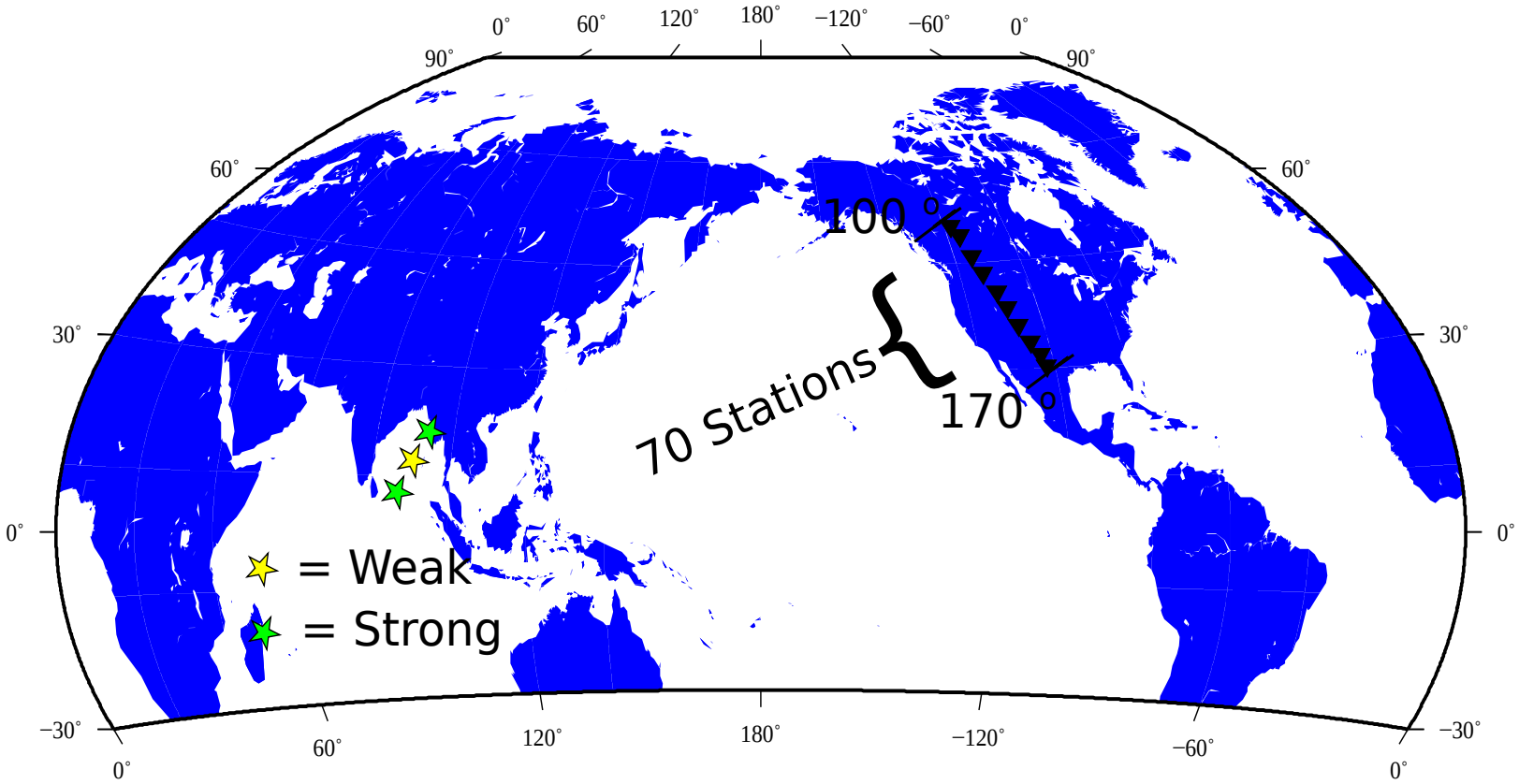
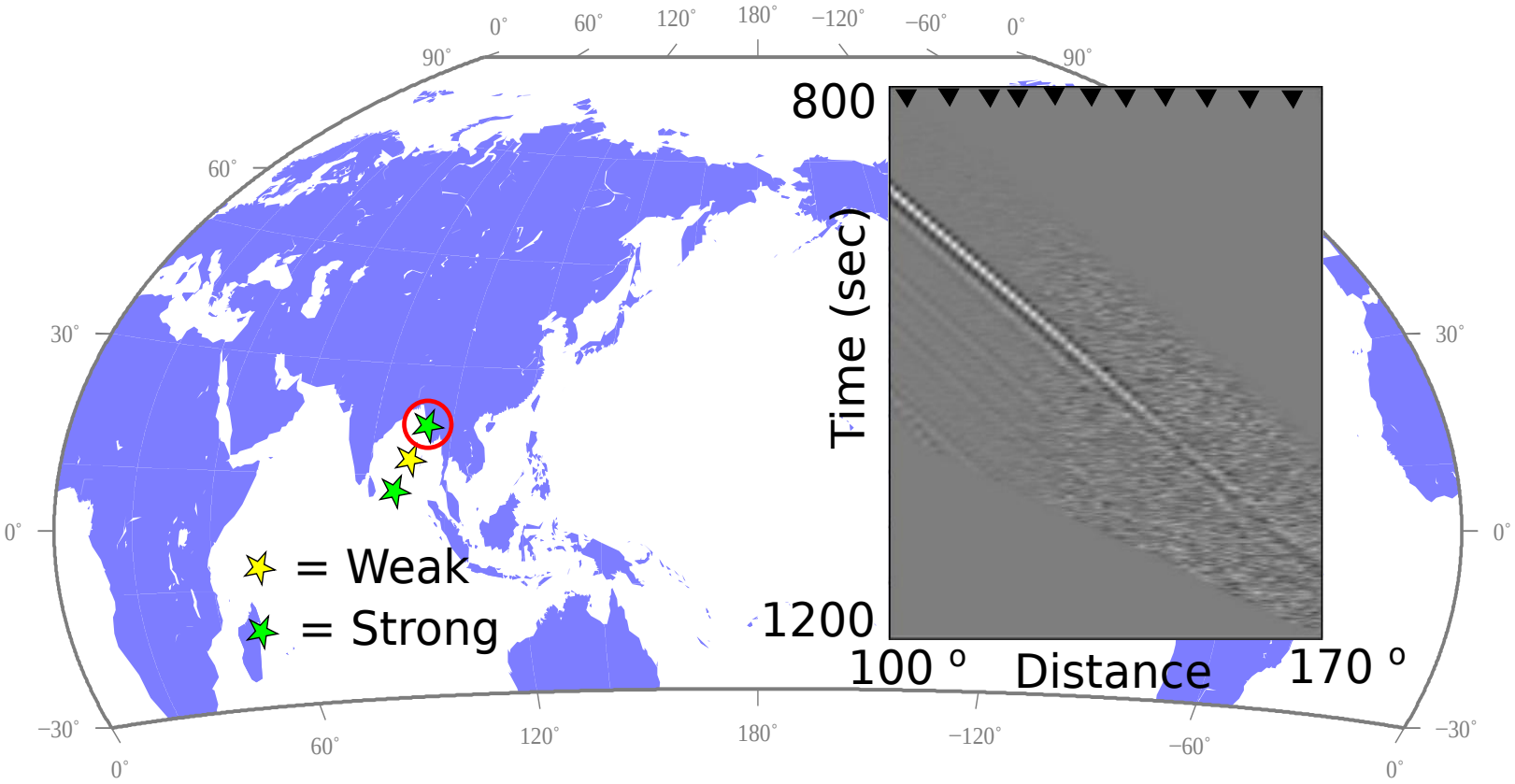
3. Conclusions

Synthetic Dataset



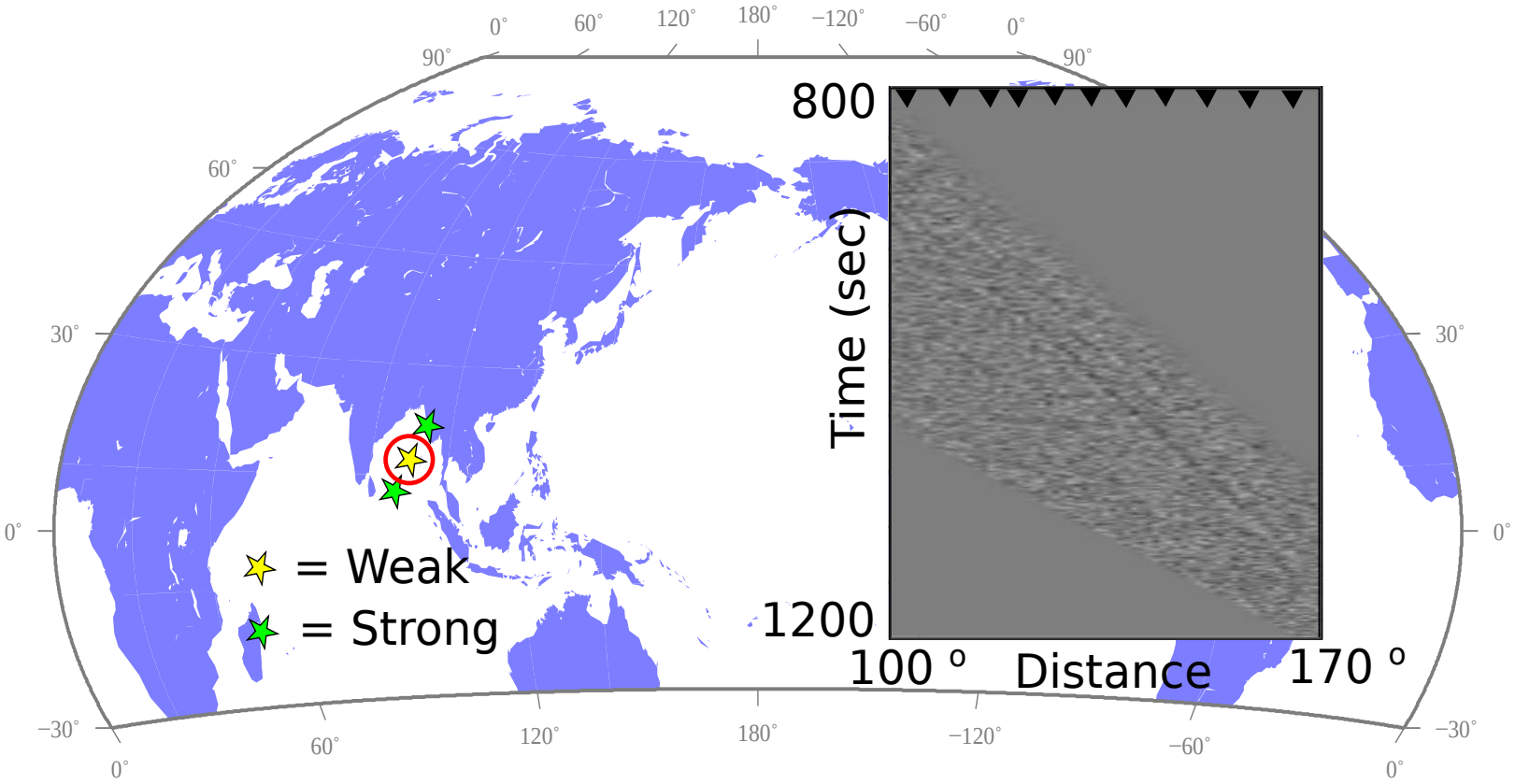
Z-component records of 2 Strong Eqs. and 1 Weak Eq.

Synthetic Data; Strong Eq. Records



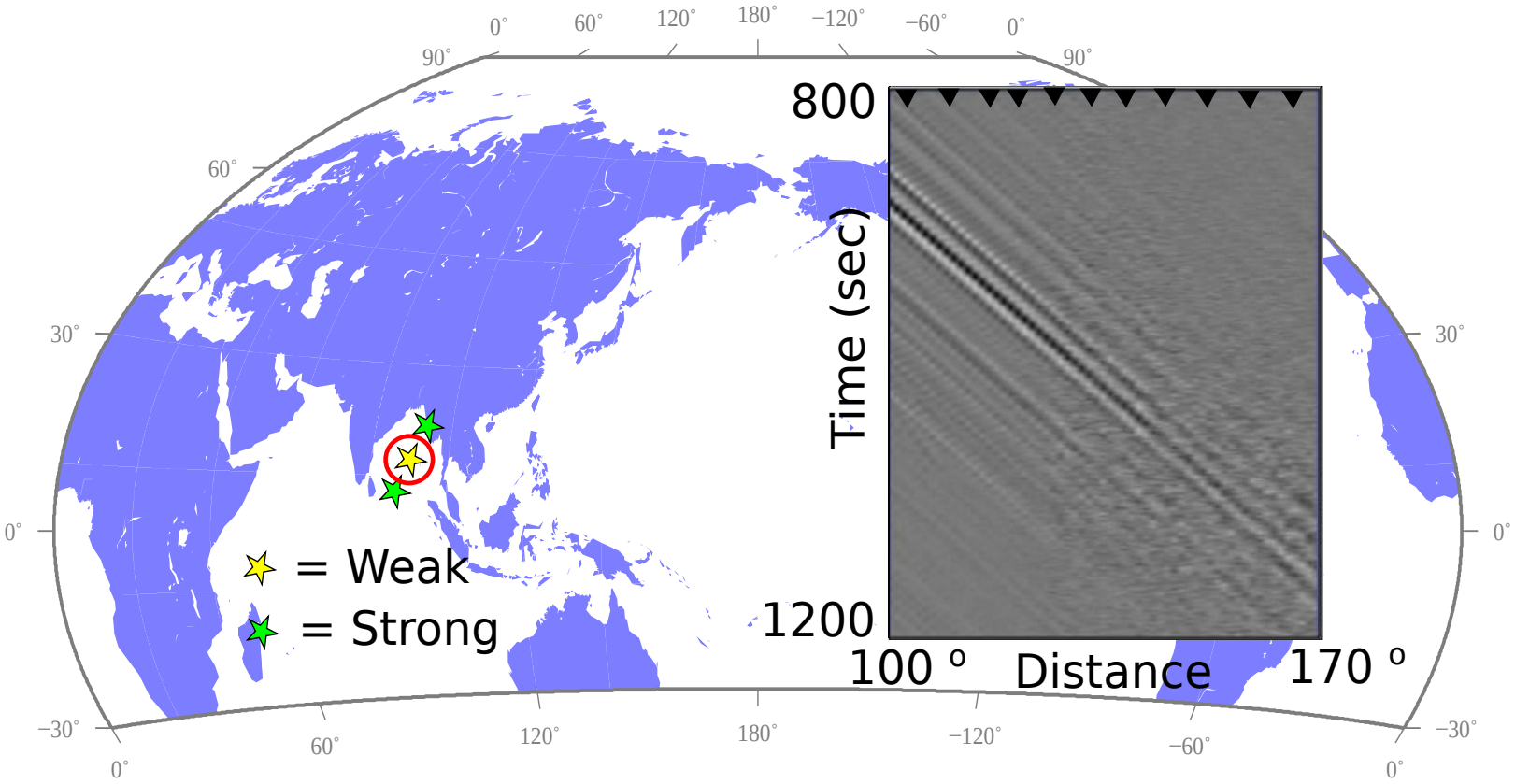
Added 0.001-0.12 Hz white noise

Synthetic Data; Weak Eq. Records



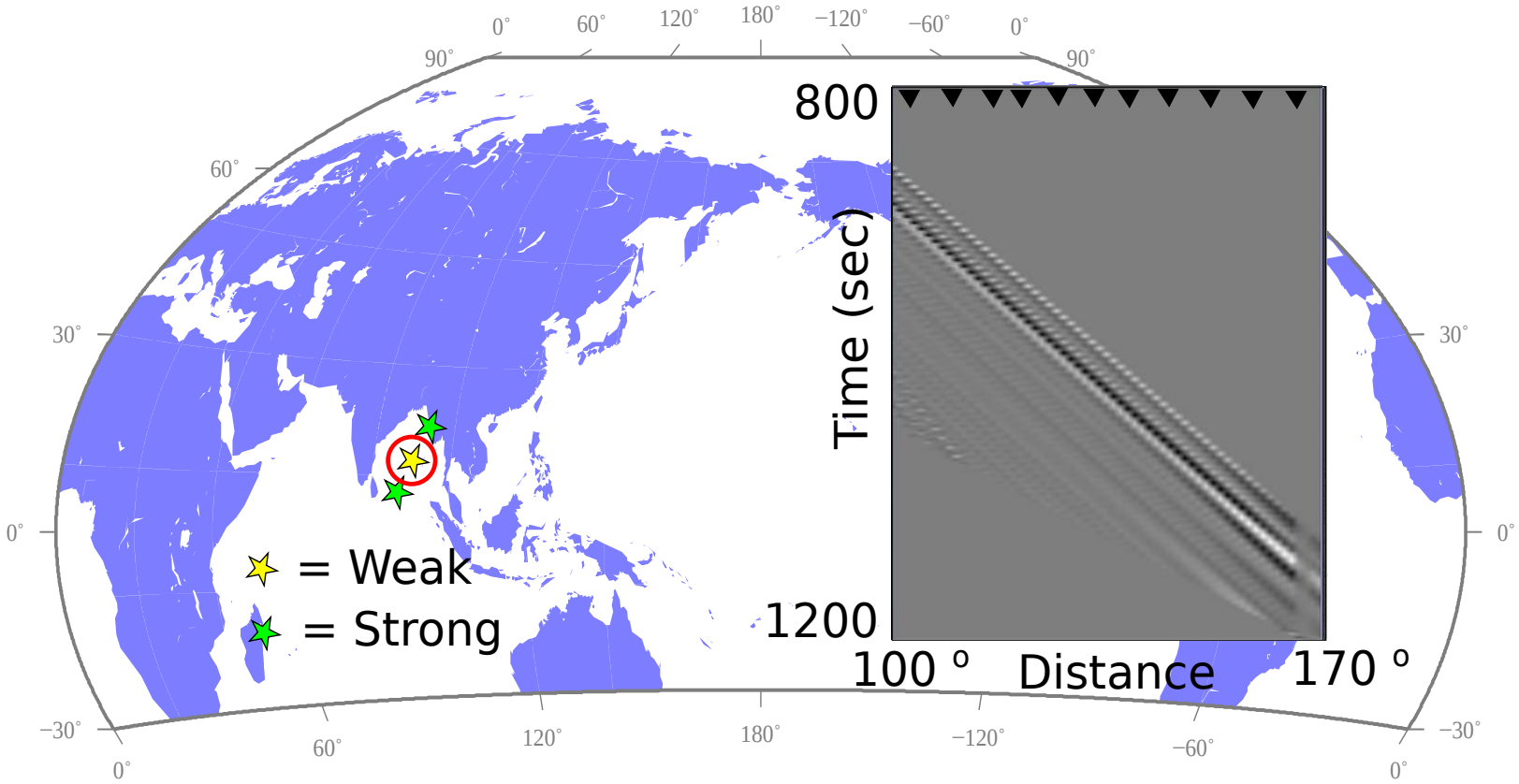
Added 0.001-0.12 Hz white noise with SNR $\approx$ 0.03

Synthetic Data; Super-virtual Weak Eq. Records



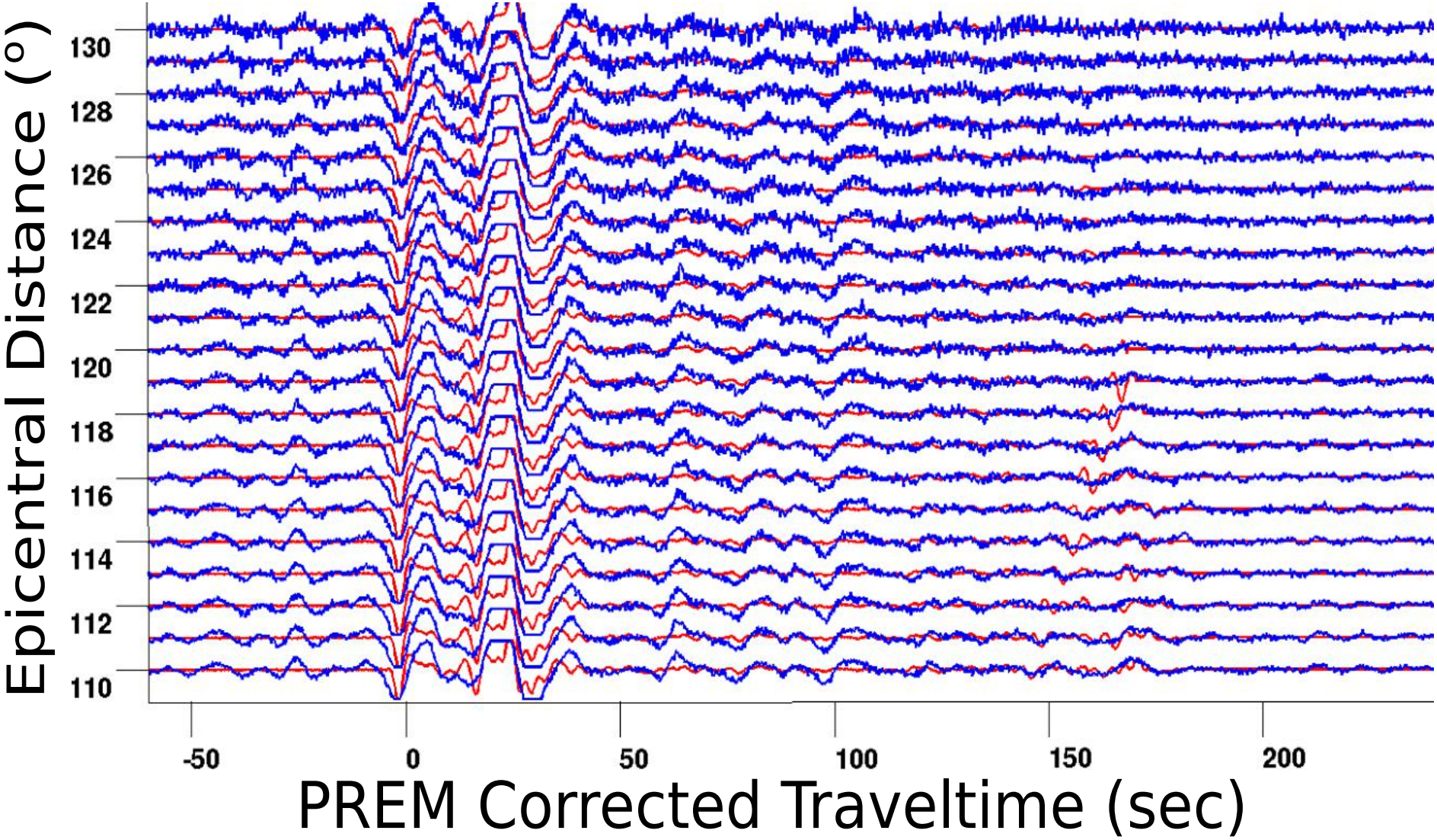
Enhanced P-diff arrival

Synthetic Data; Actual Weak Eq. Records



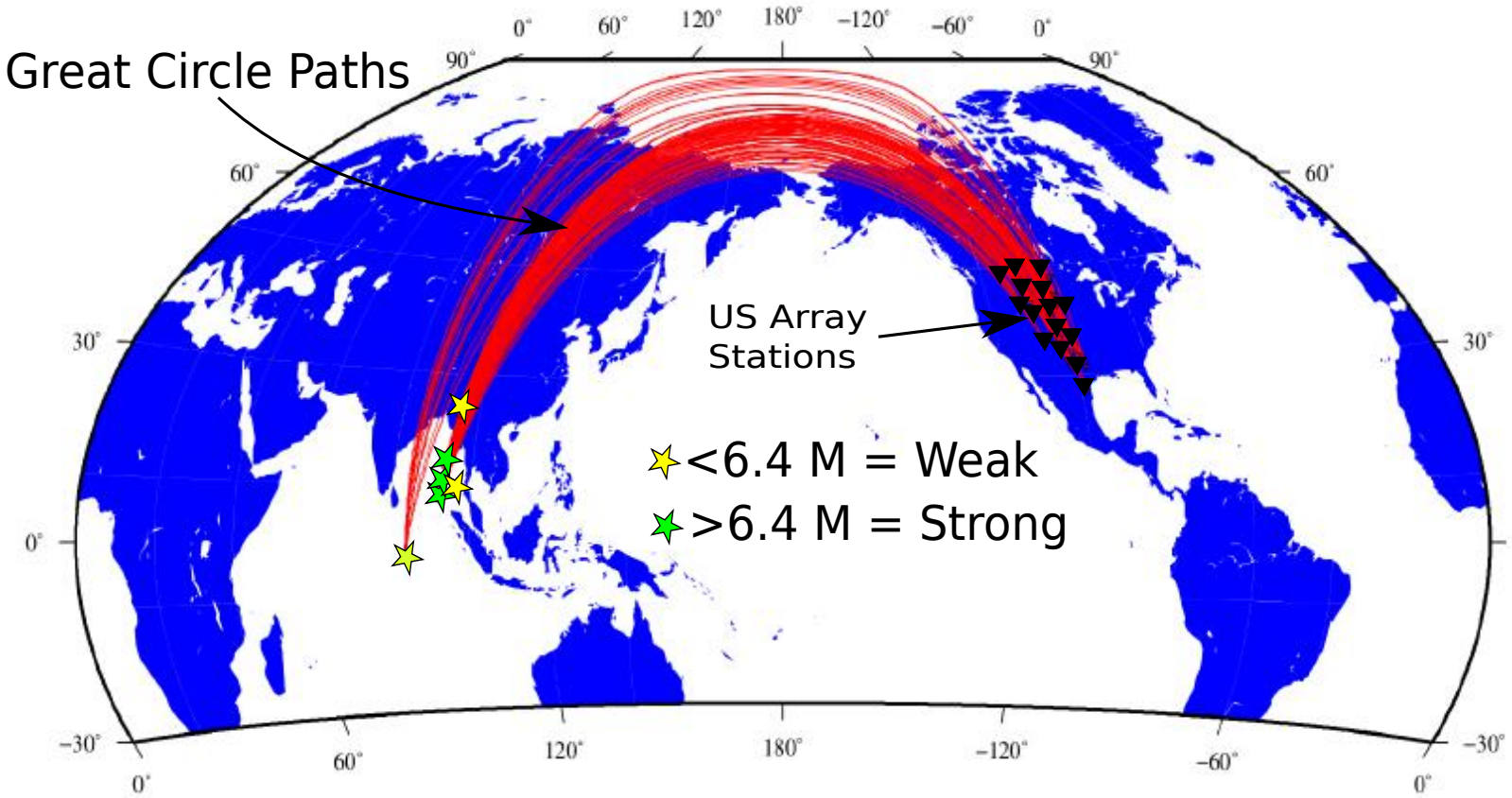
Before adding random noise

Synthetic Data; SVI Vs. Actual Waveforms



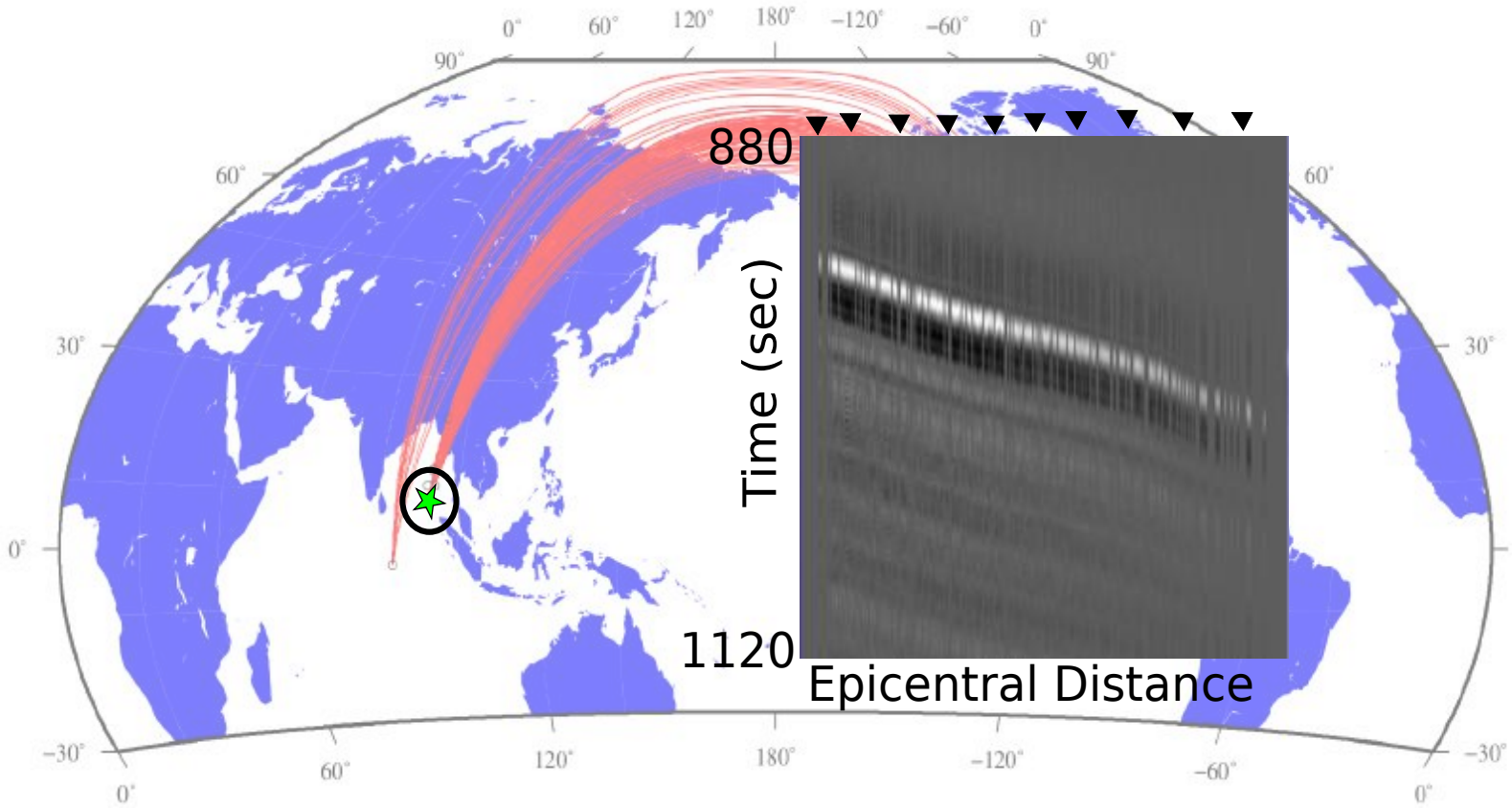
— Raw Synthetic Records — SVI Records

USArray Dataset



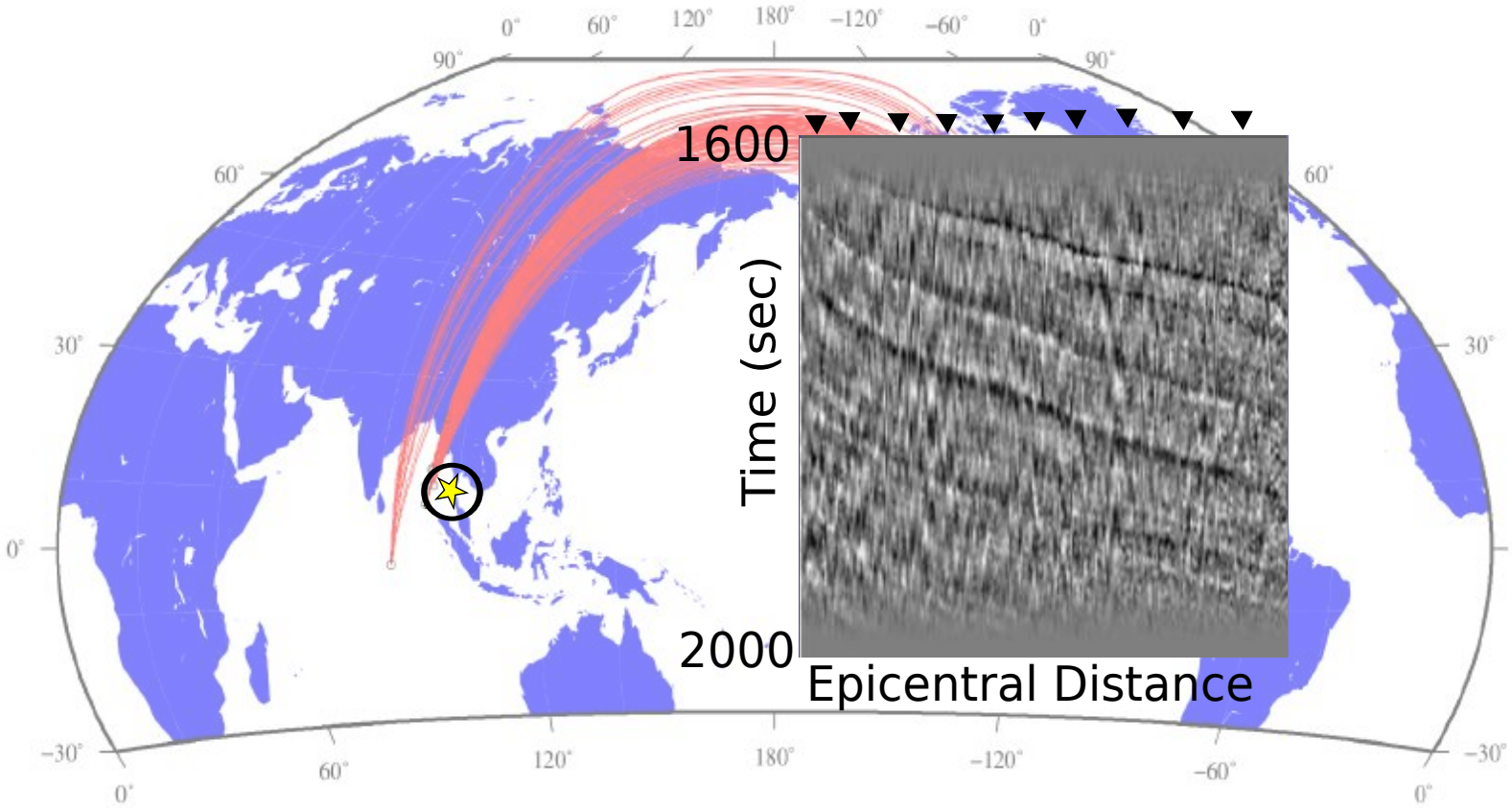
5 Earthquakes near Indonesia on approx. same 2D plane

USArray Dataset; Strong Eq. Records



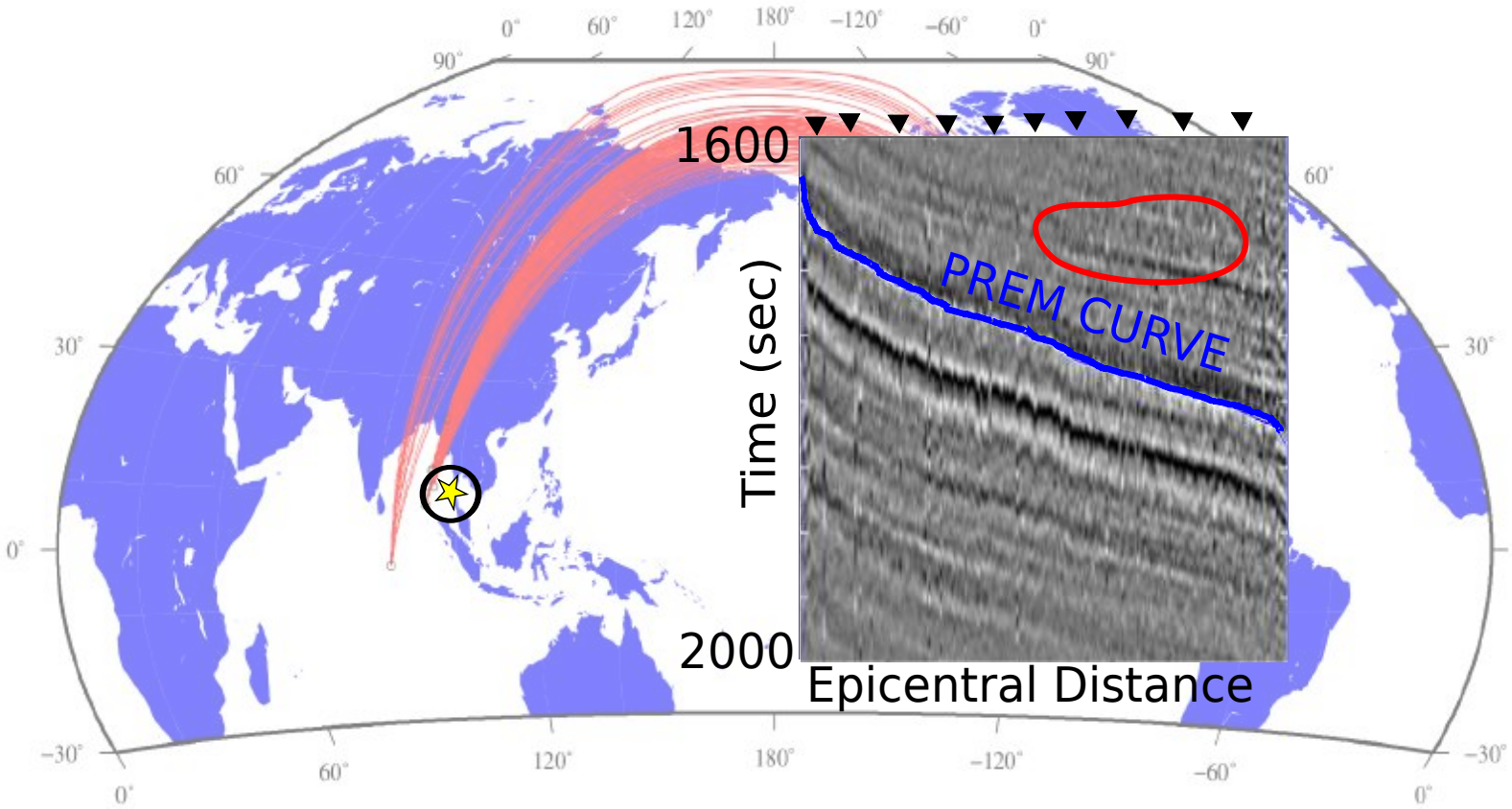
Strong event records used while processing with high SNR

USArray Dataset; 31-May-2010 Eq. Records



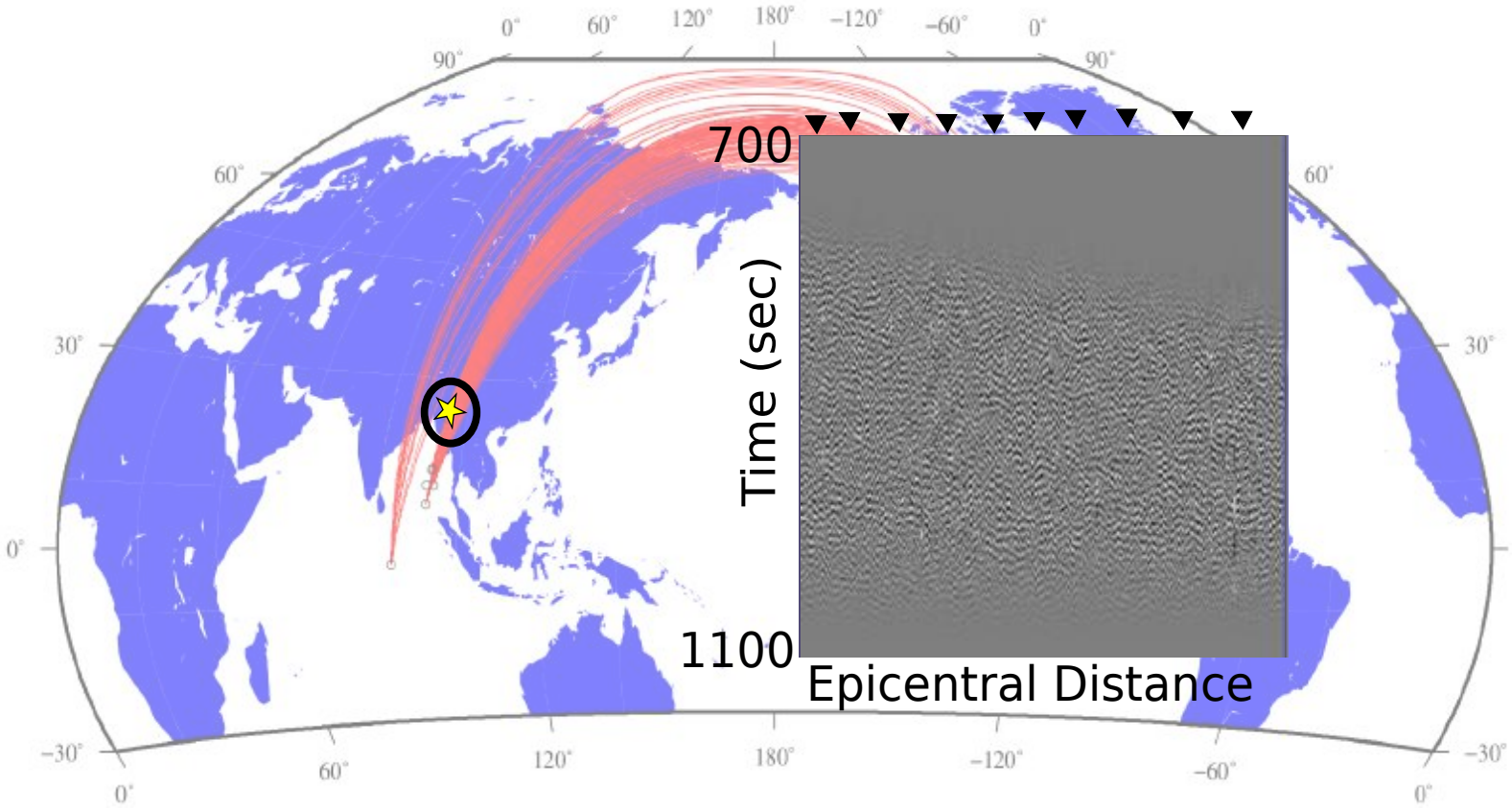
Intermediate magnitude (6.4 M) T-component records

USArray Dataset; 31-May-2010 Eq. Records



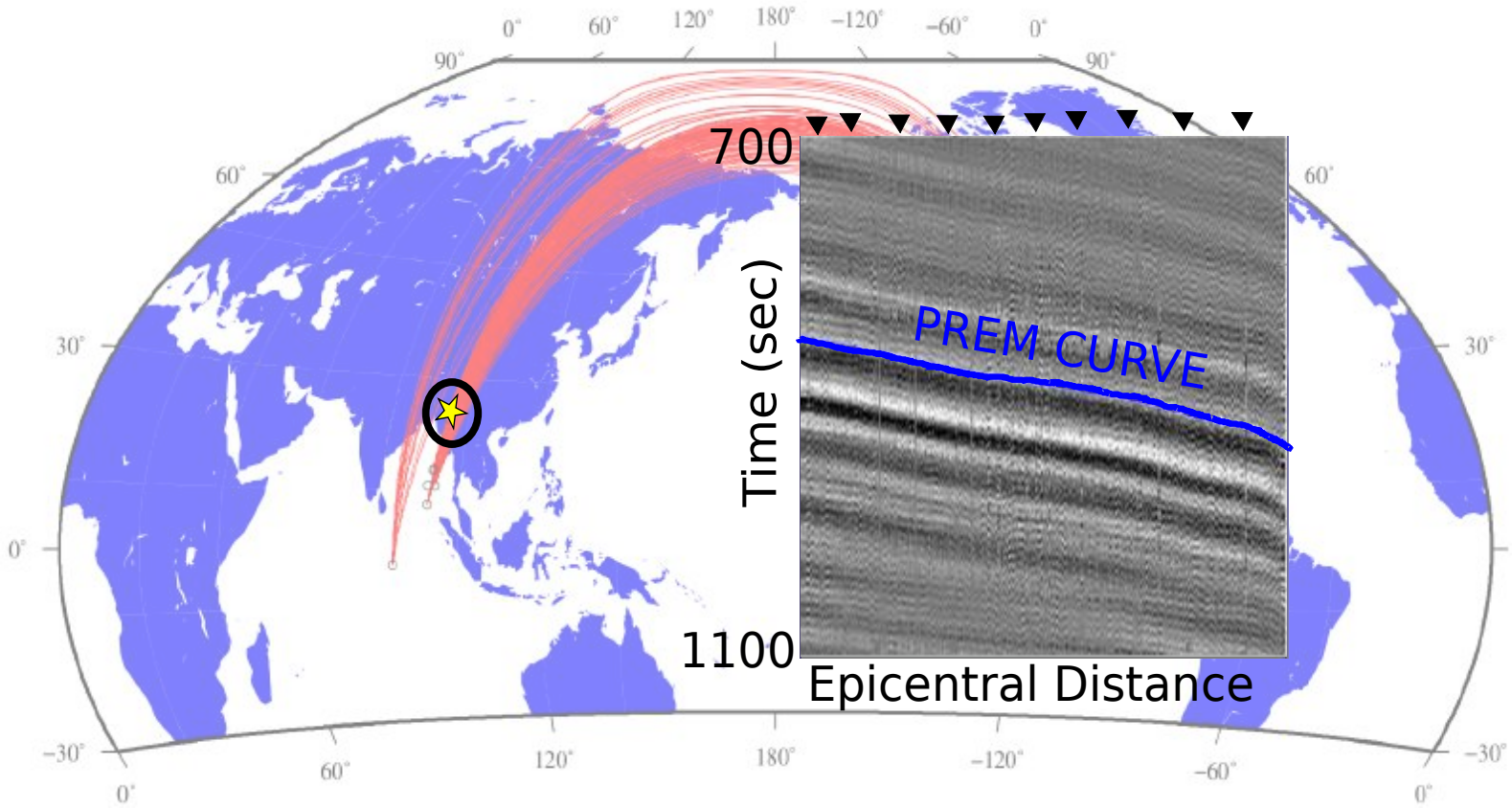
SVI records – other arrivals except S-diff are suppressed

USArray Dataset; 12-Mar-2010 Eq. Records



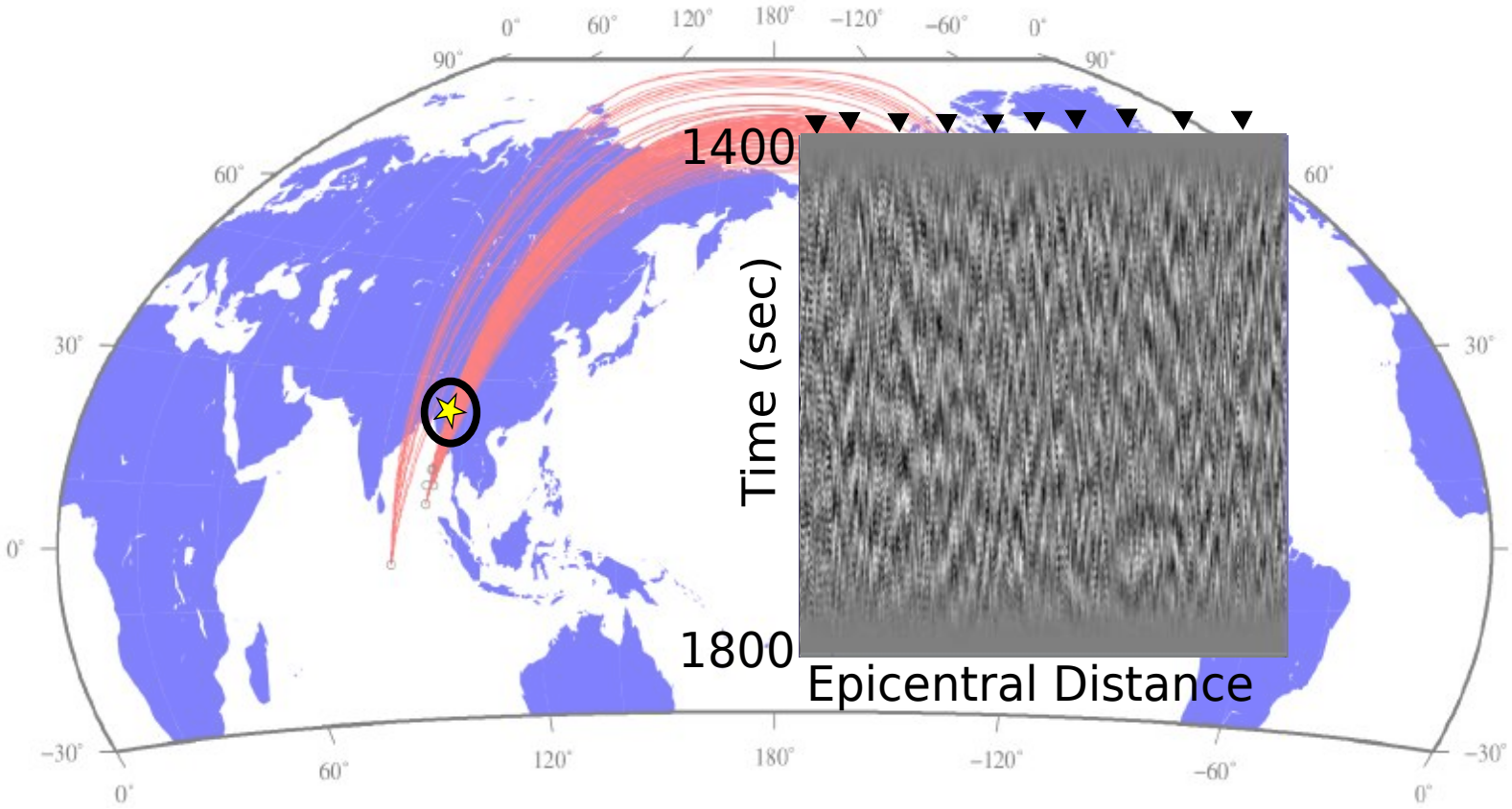
Low magnitude (5.5 M) Z-component records

USArray Dataset; 12-Mar-2010 Eq. Records



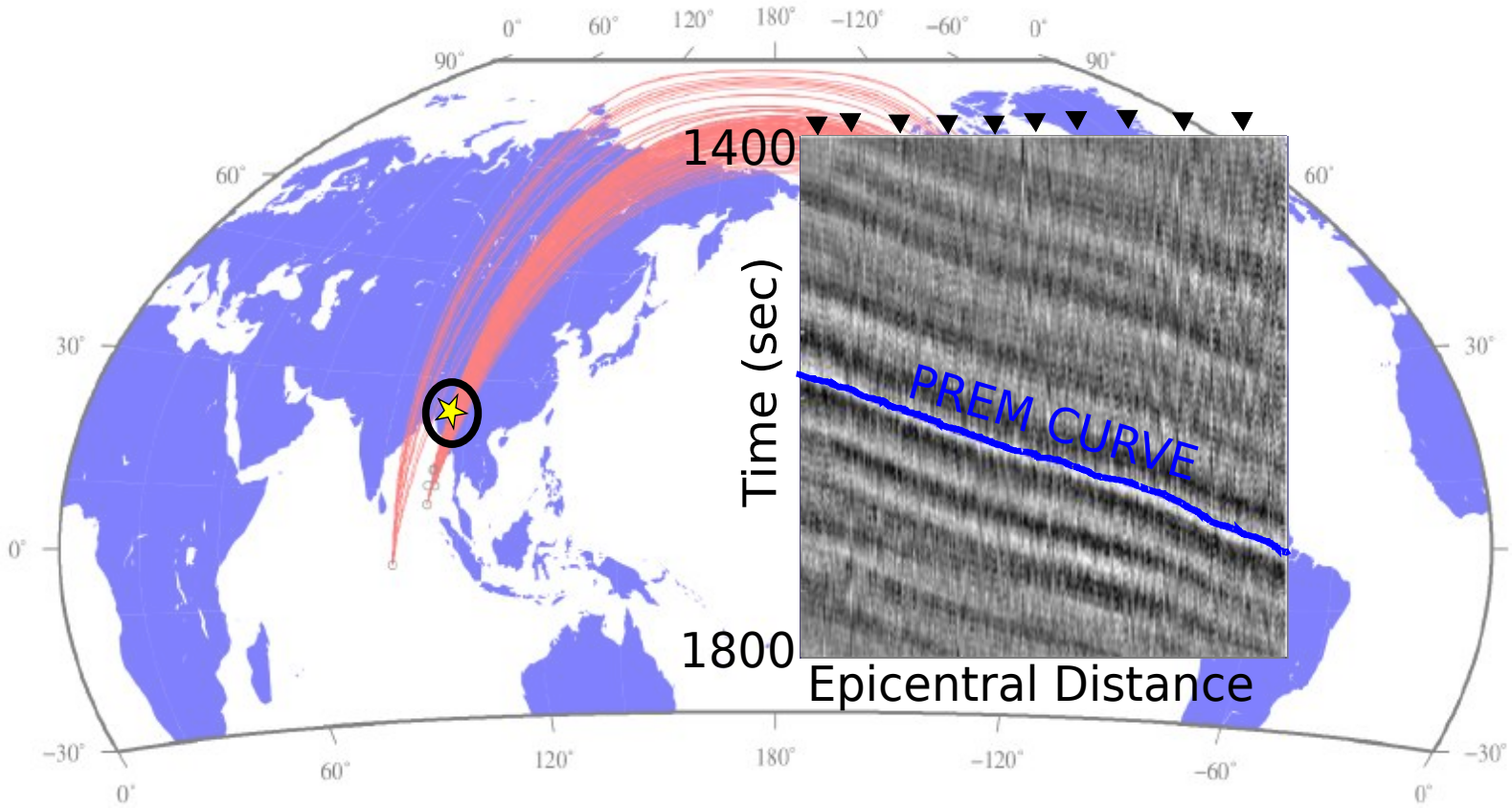
SVI records showing P-diff arrivals

USArray Dataset; 12-Mar-2010 Eq. Records



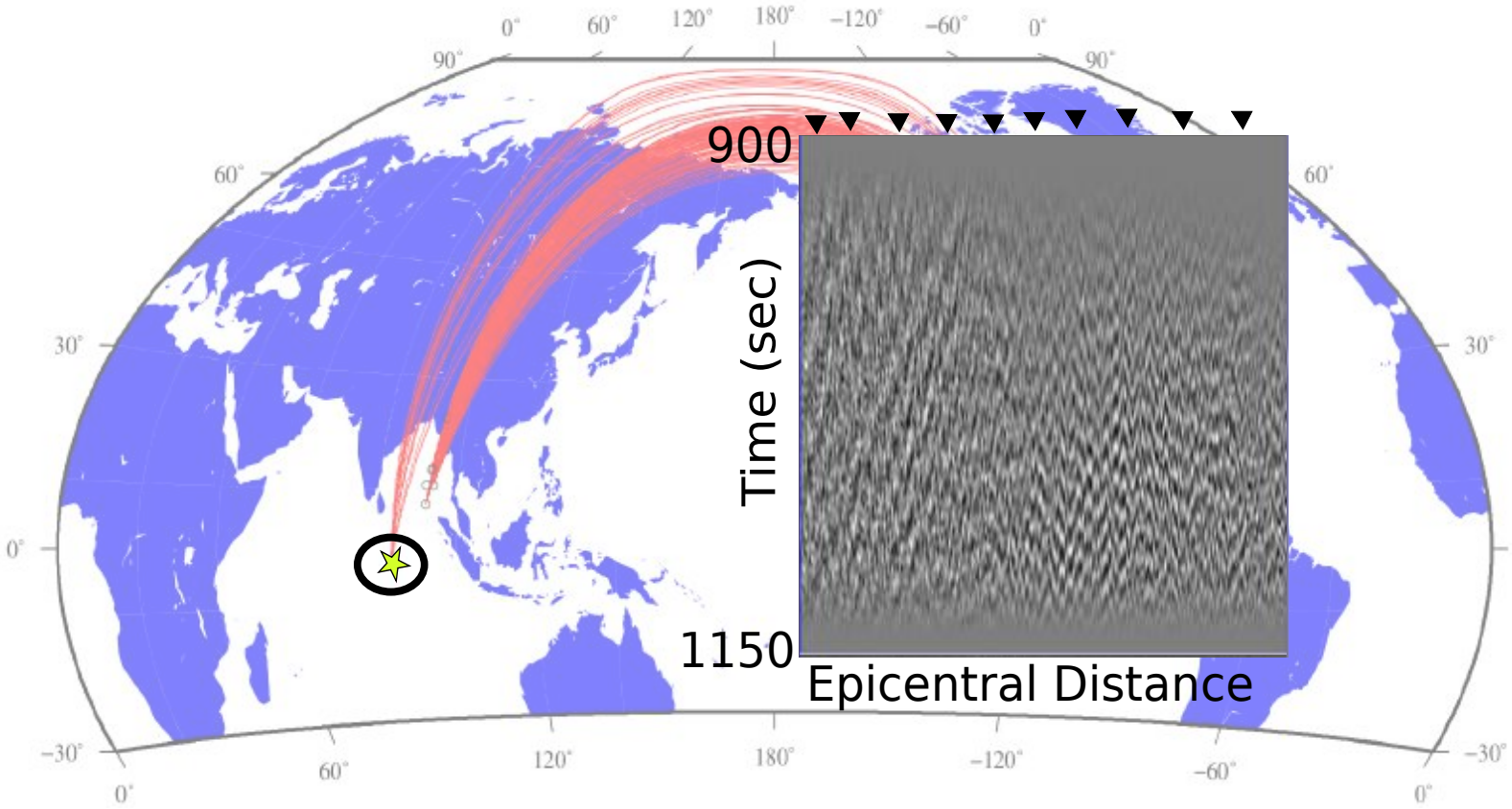
Low magnitude (5.5 M) T-component records

USArray Dataset; 12-Mar-2010 Eq. Records



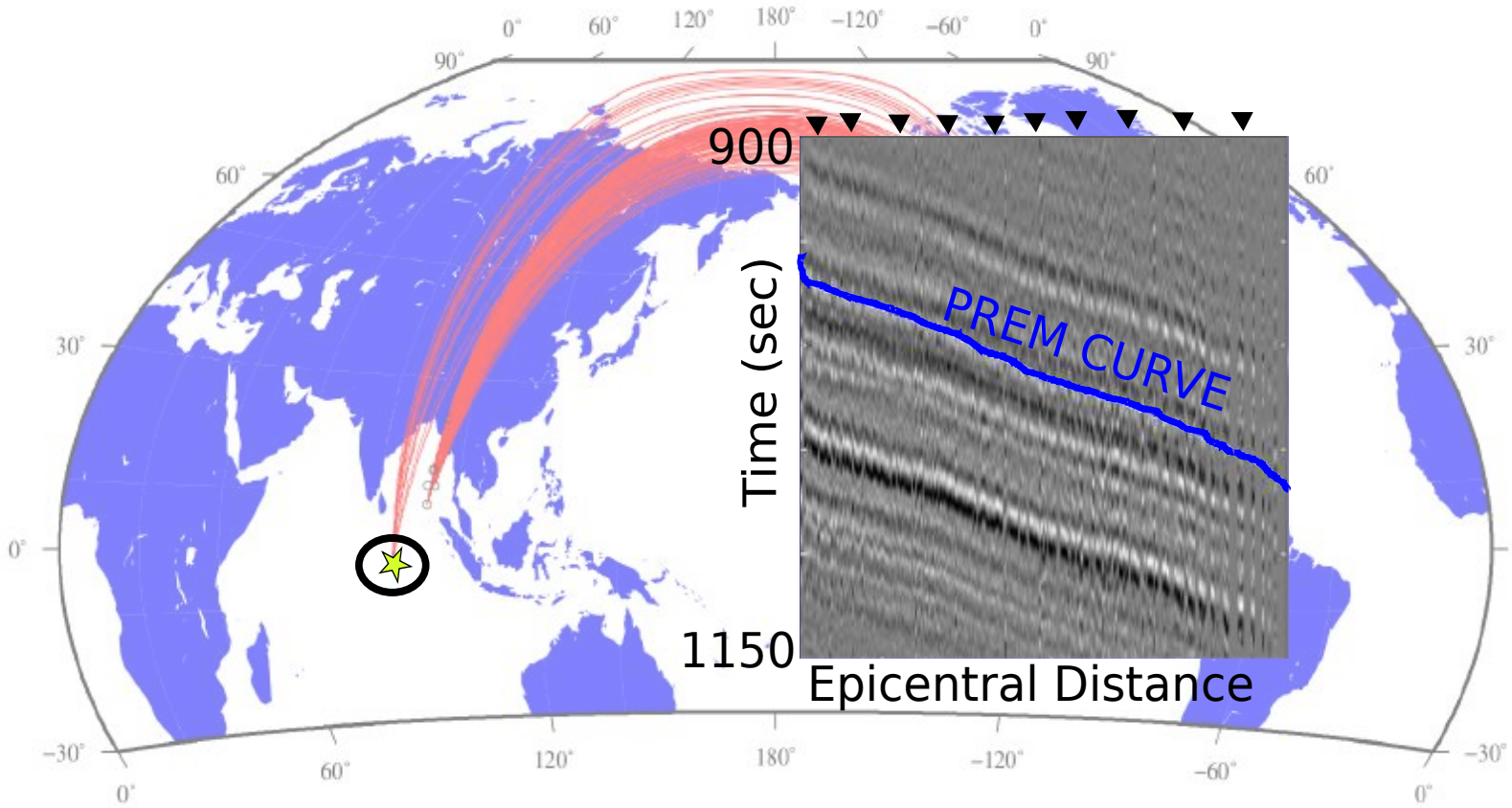
SVI records showing S-diff arrivals

USArray Dataset; 14-Mar-2010 Eq. Records



Low Magnitude (6.0 M) Z-component records

USArray Dataset; 14-Mar-2010 Eq. Records



SVI records showing S-diff arrivals

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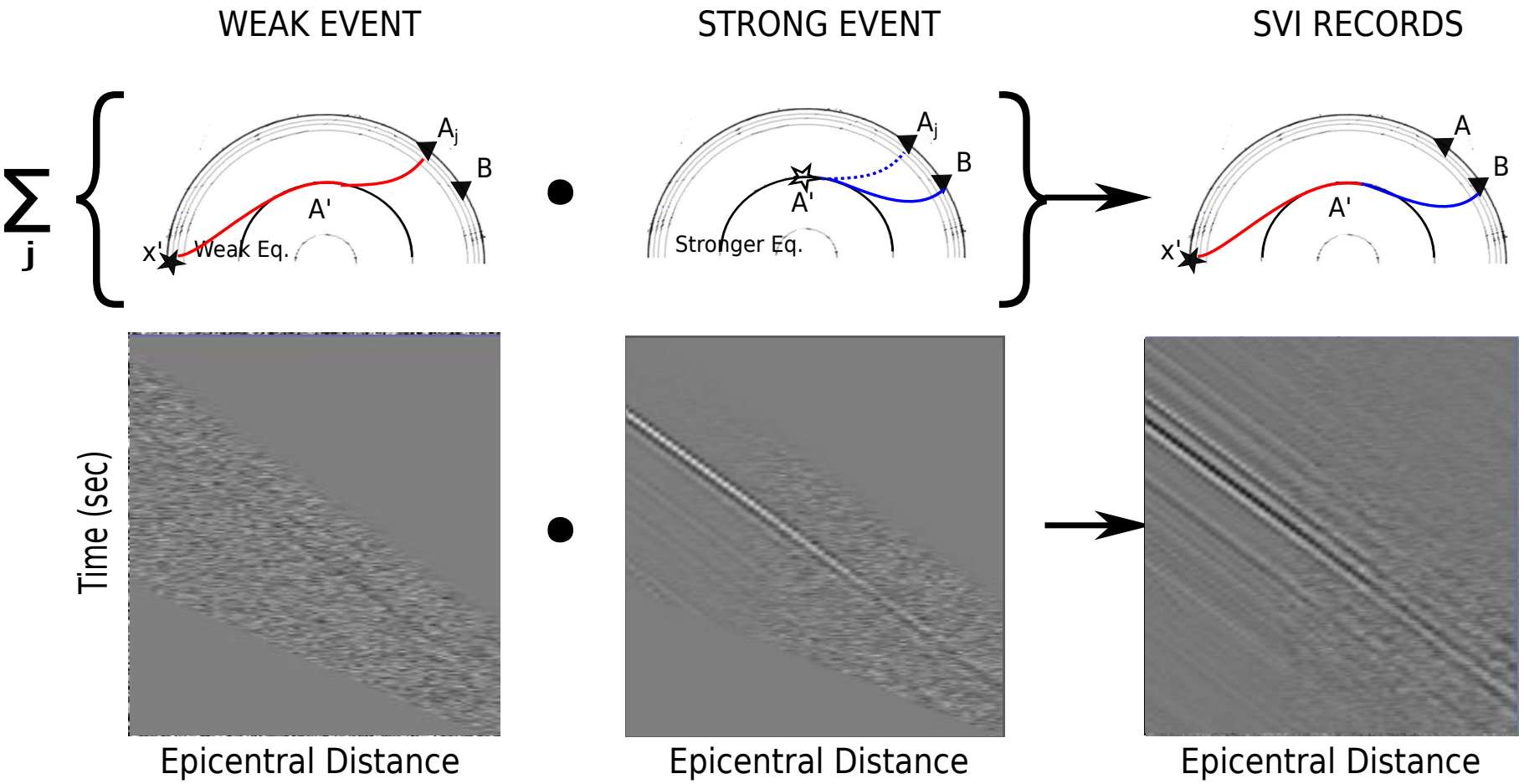
Synthetic Test



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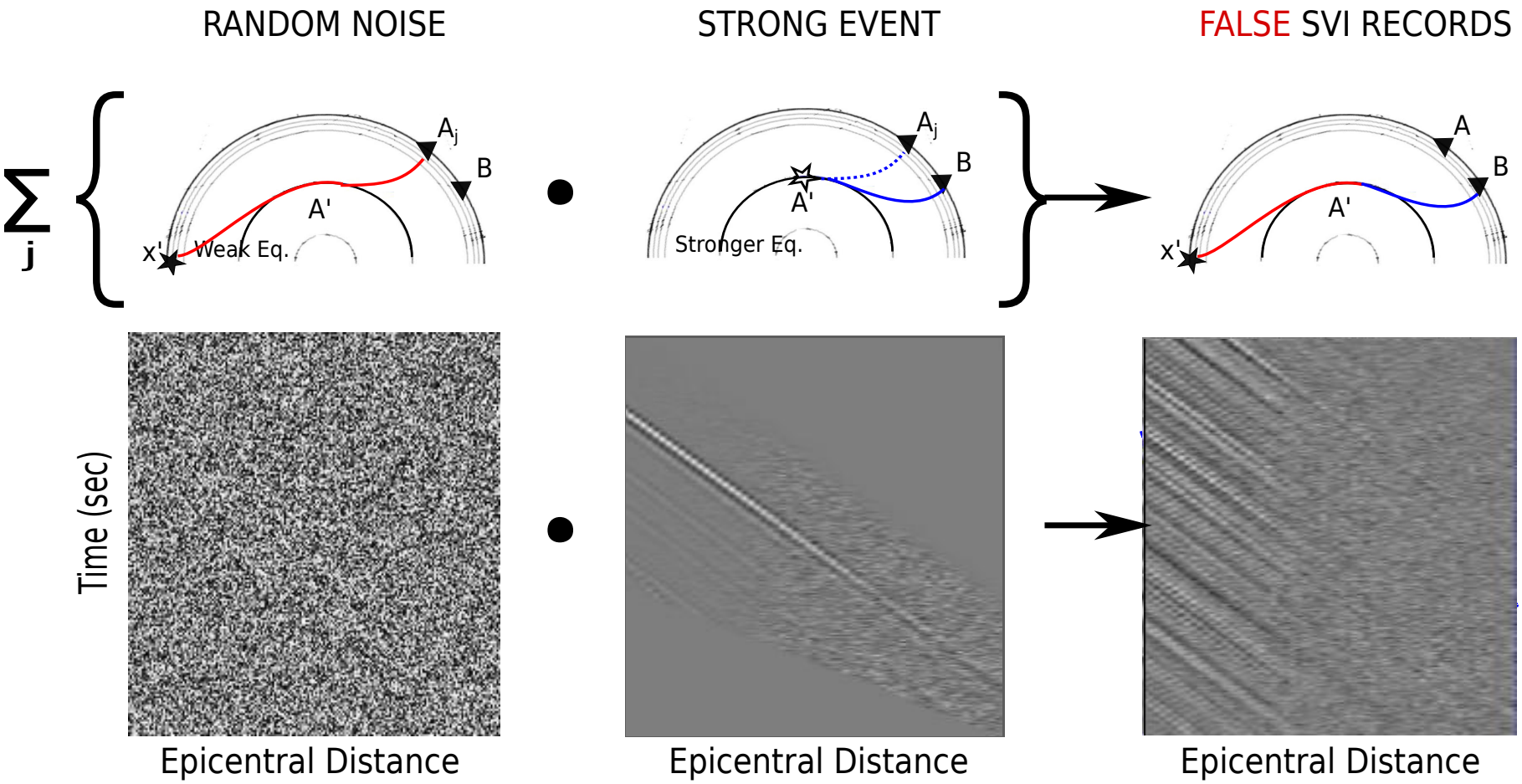
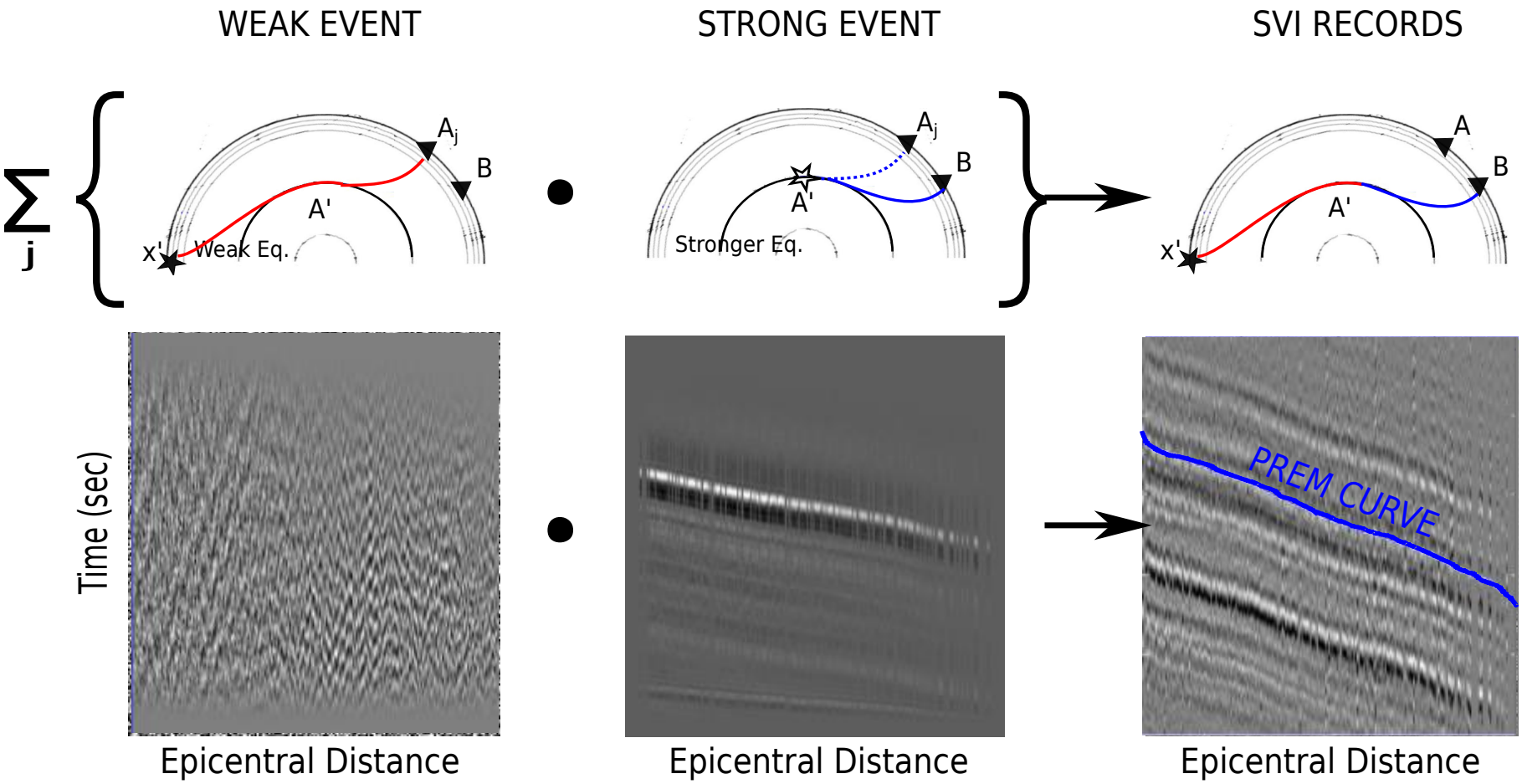


USArray Test



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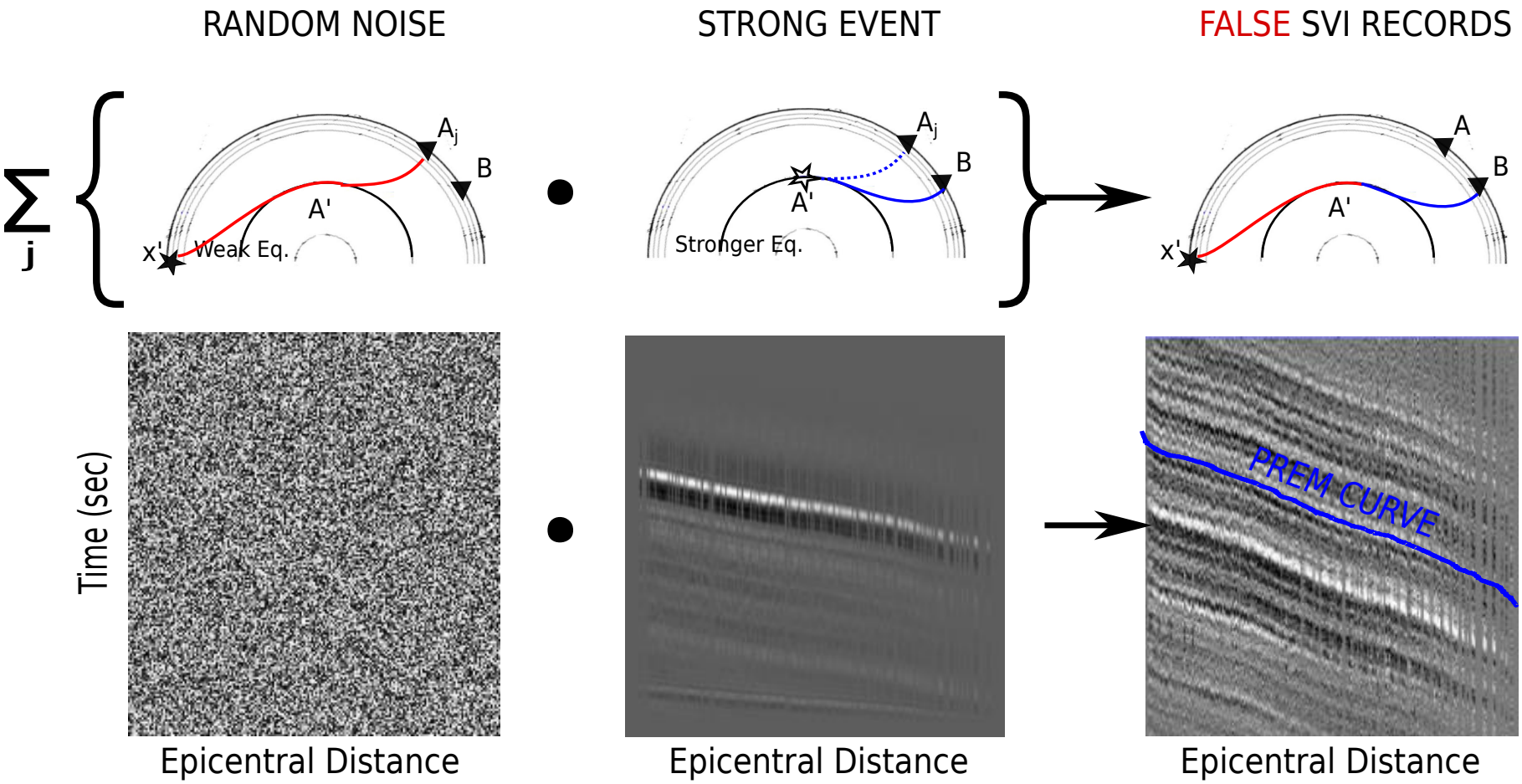

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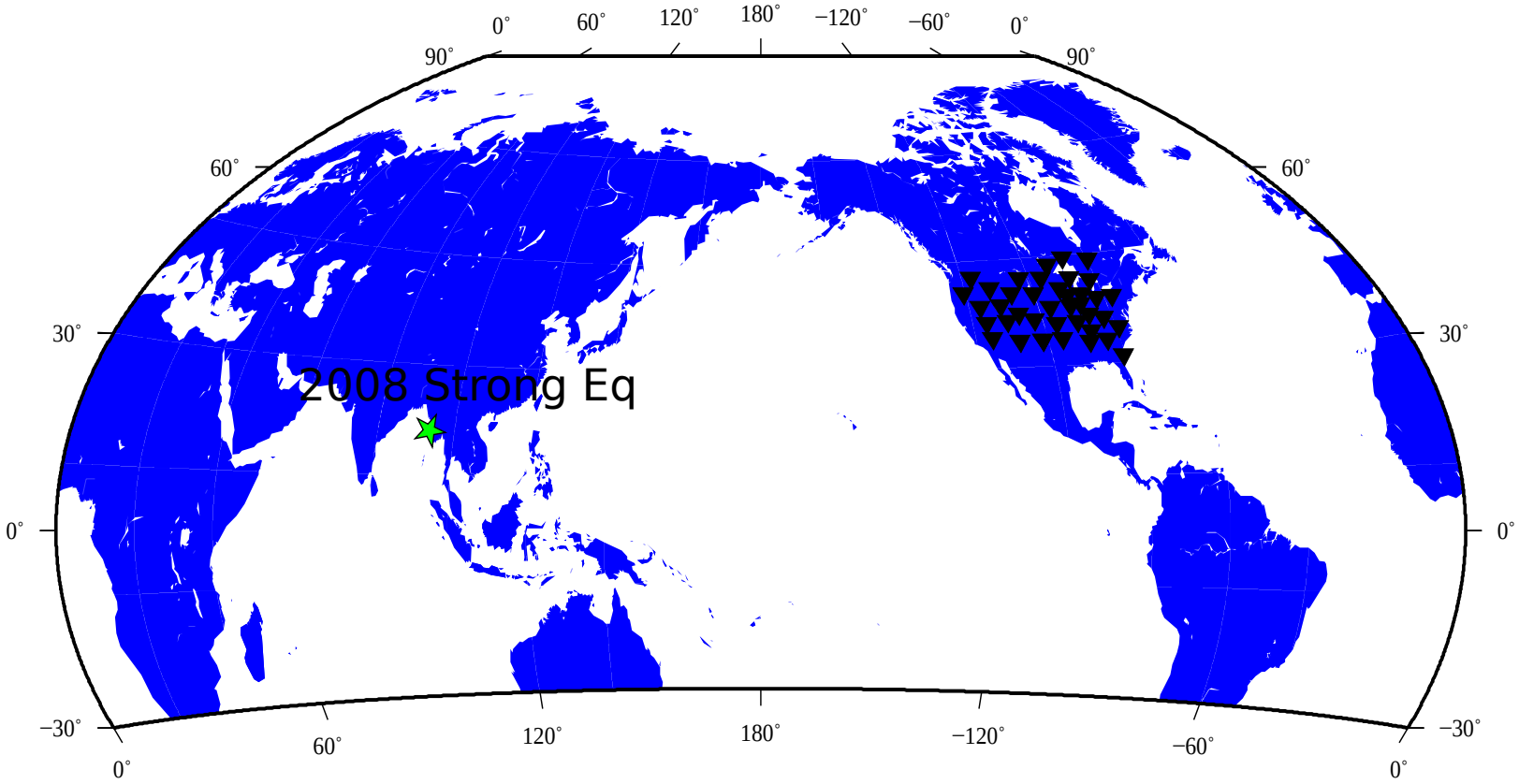
Create New Virtual Diffraction Records



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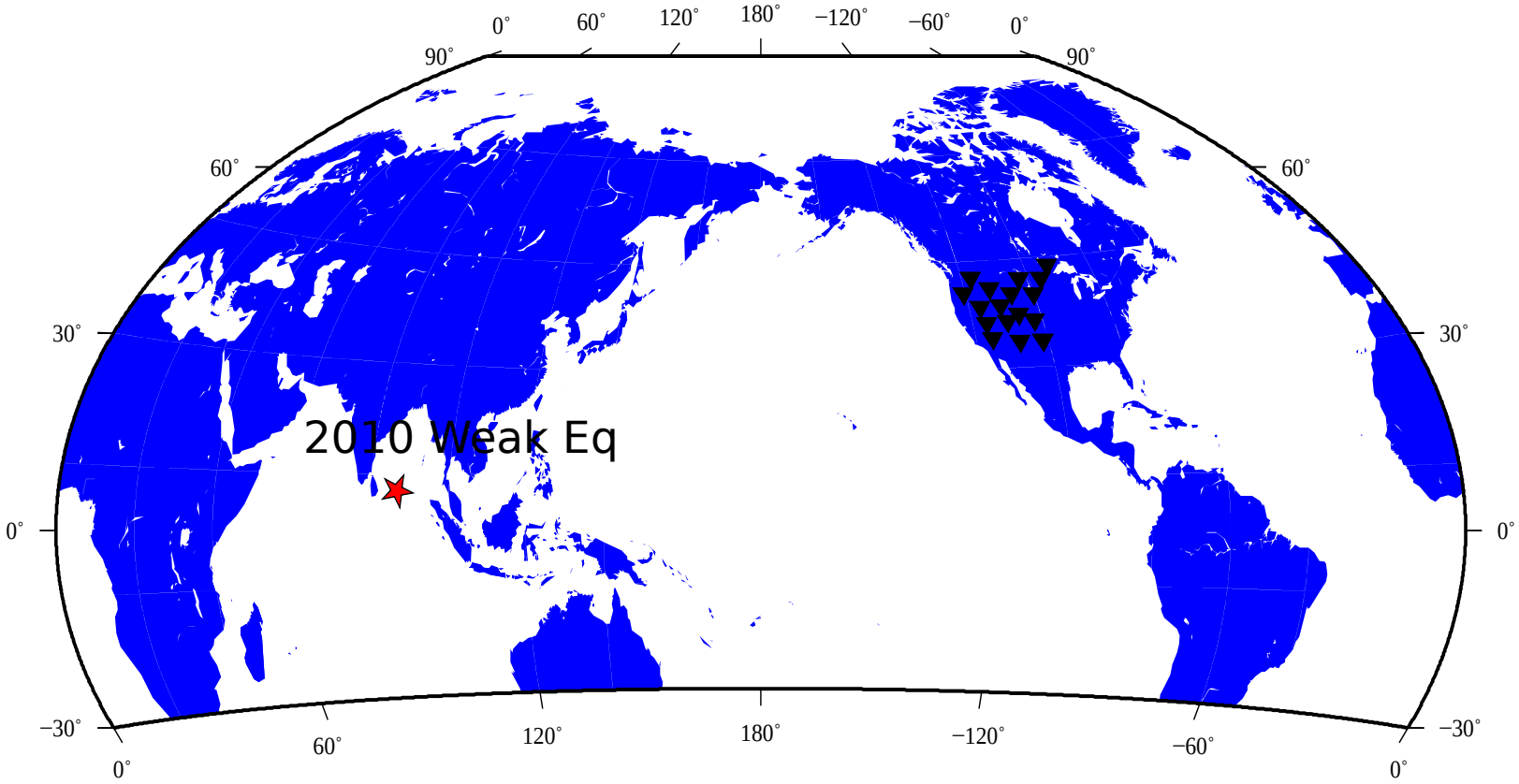

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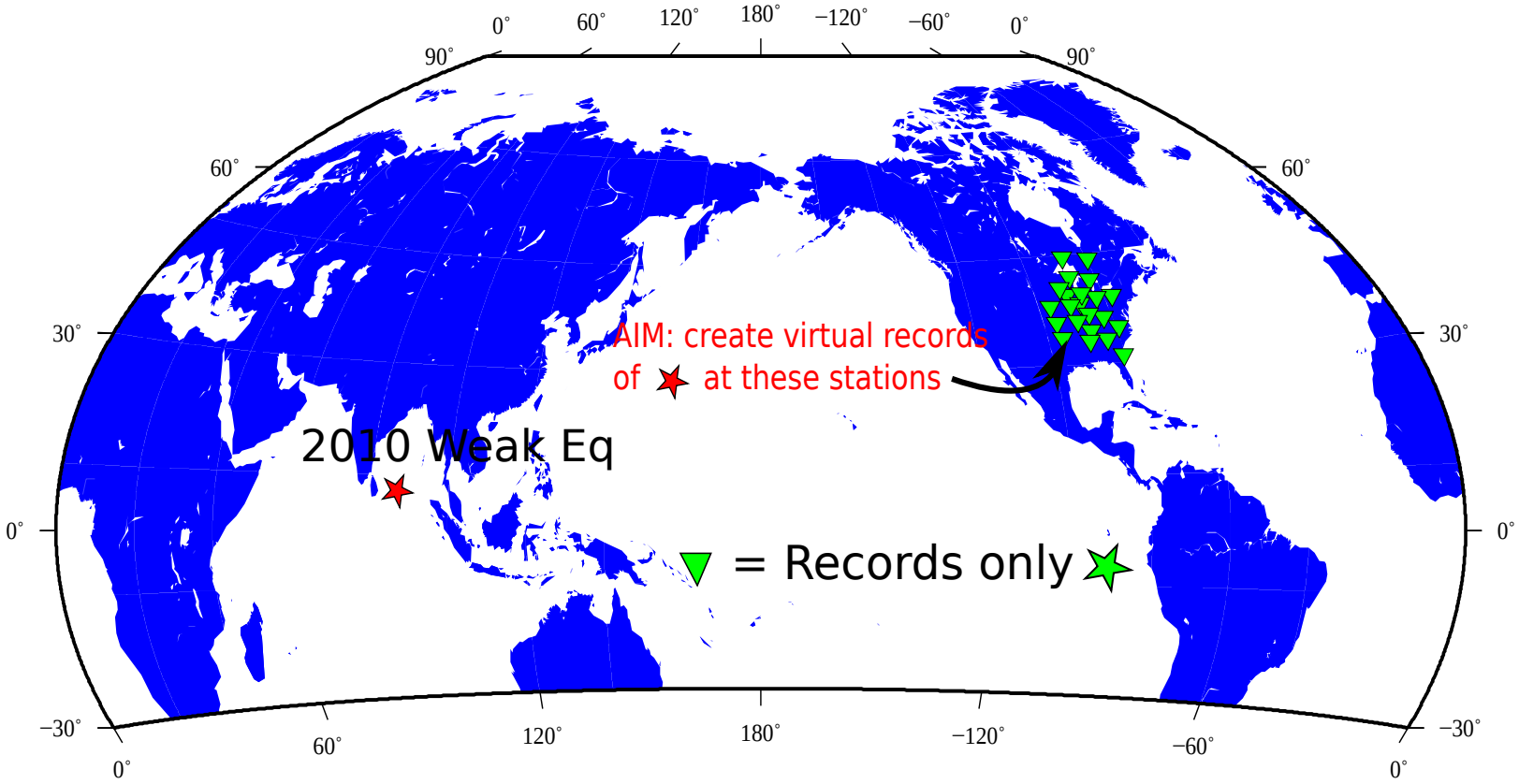

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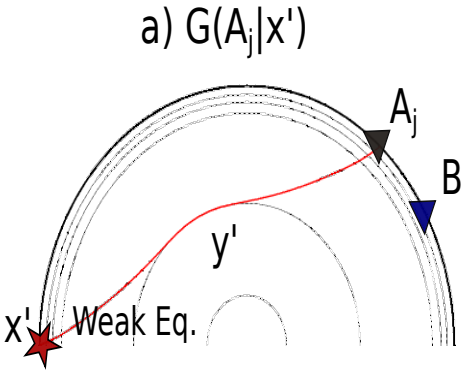
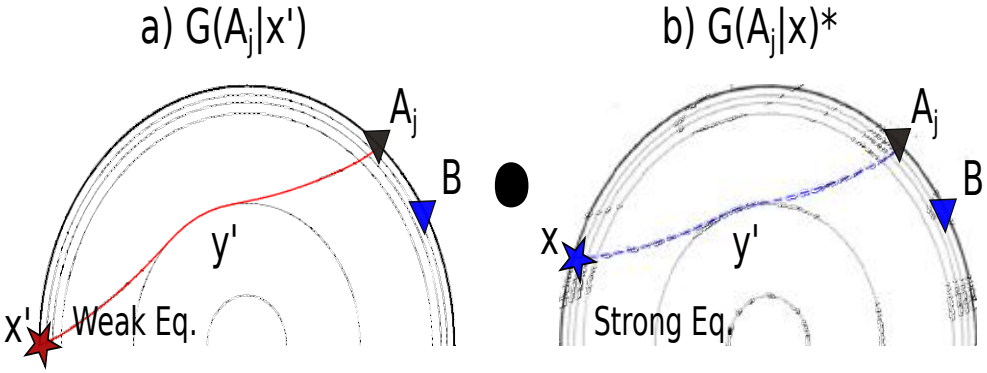
Cross-correlation



★ event recorded only at blue stations

Cross-correlation



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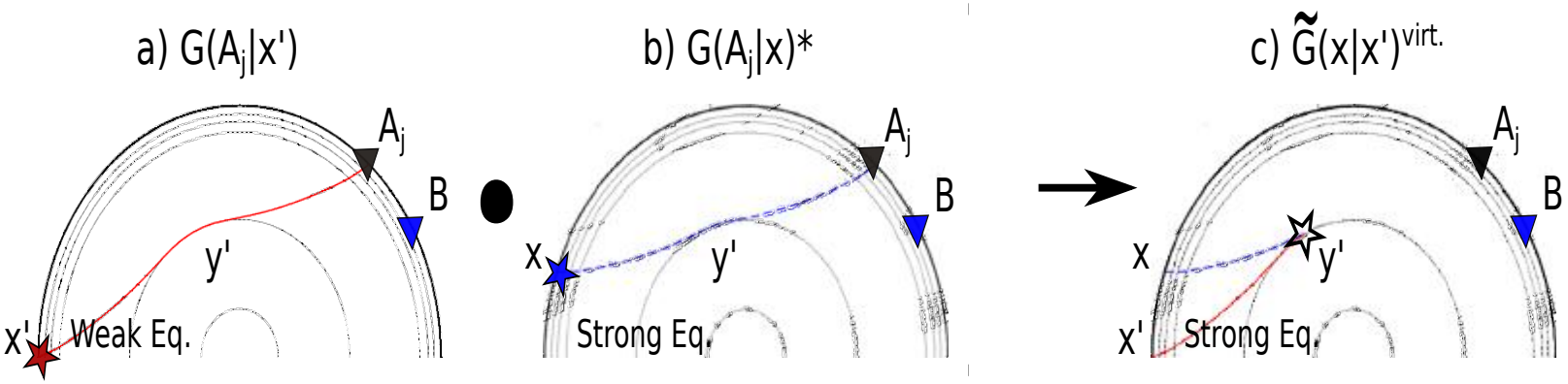

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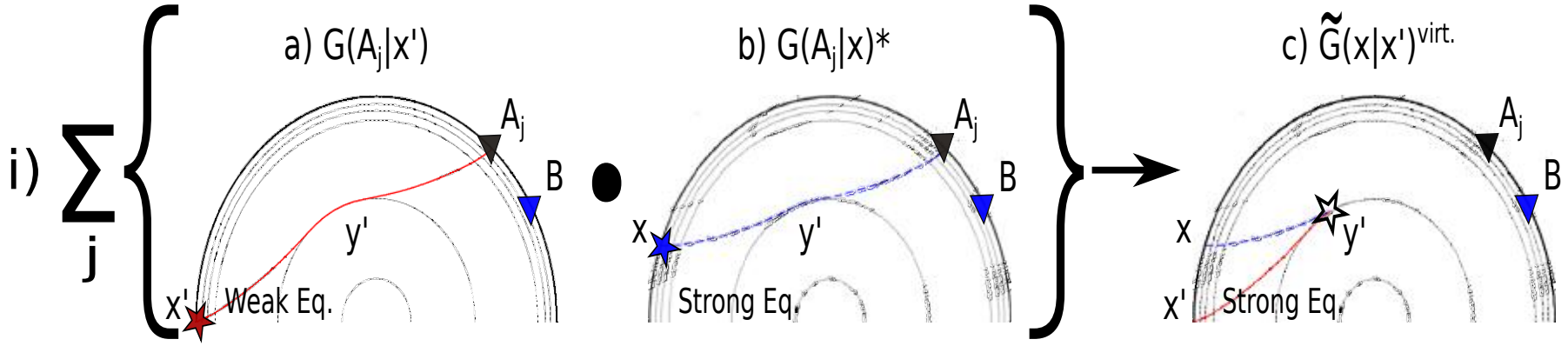

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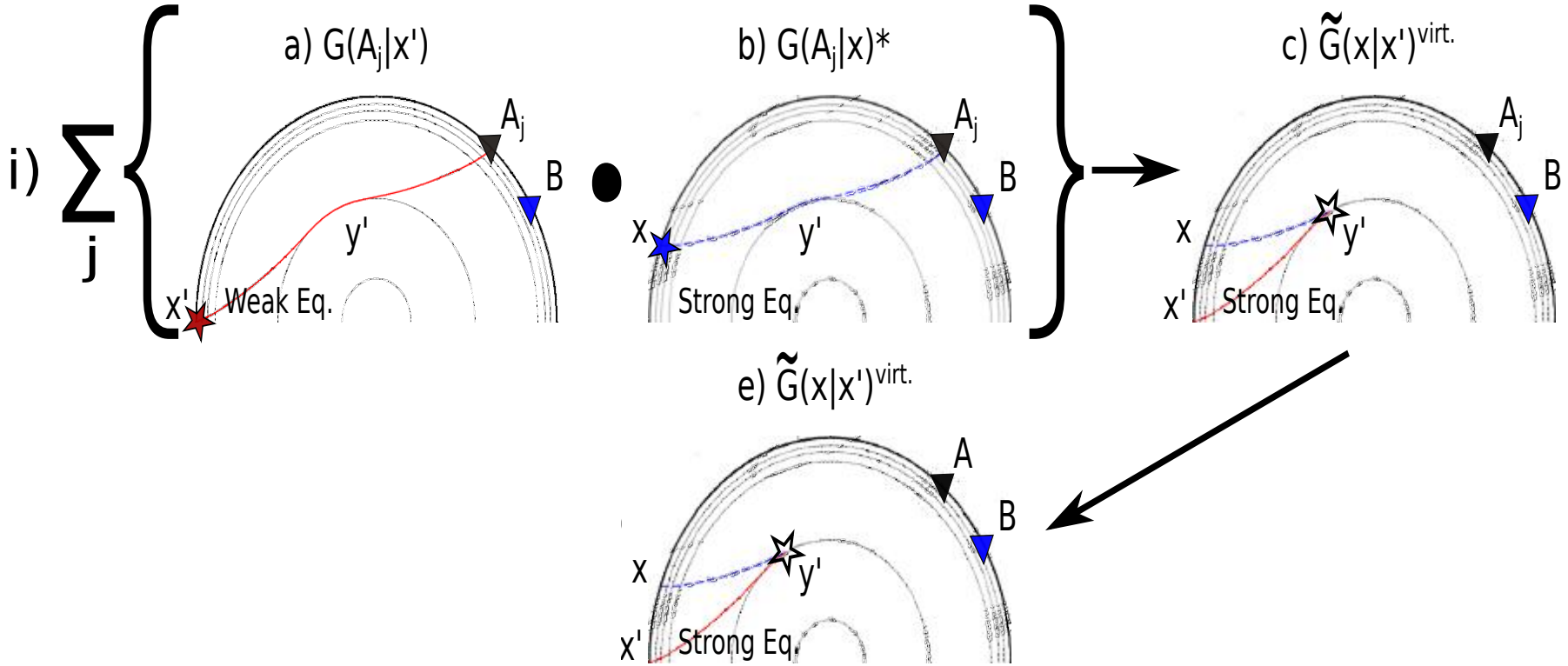
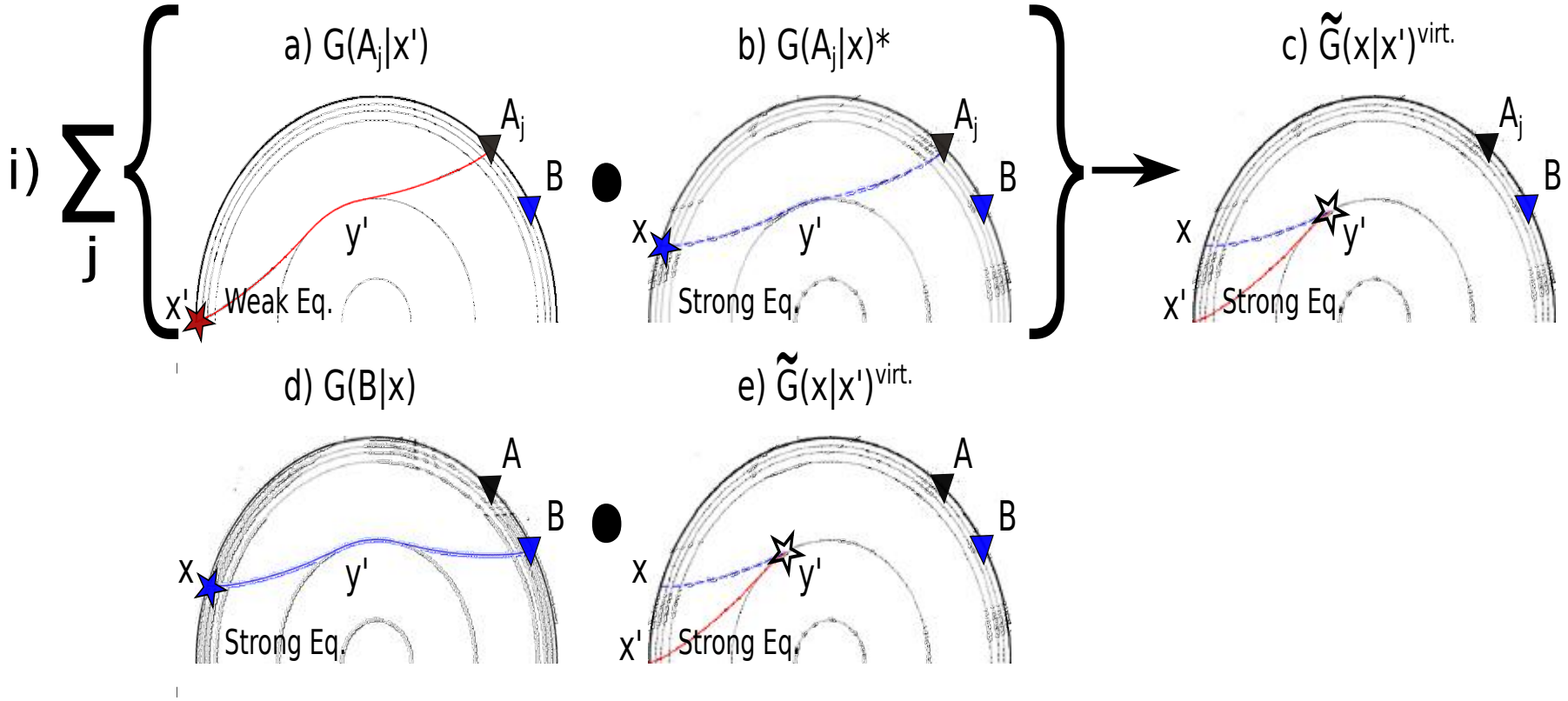
Convolution



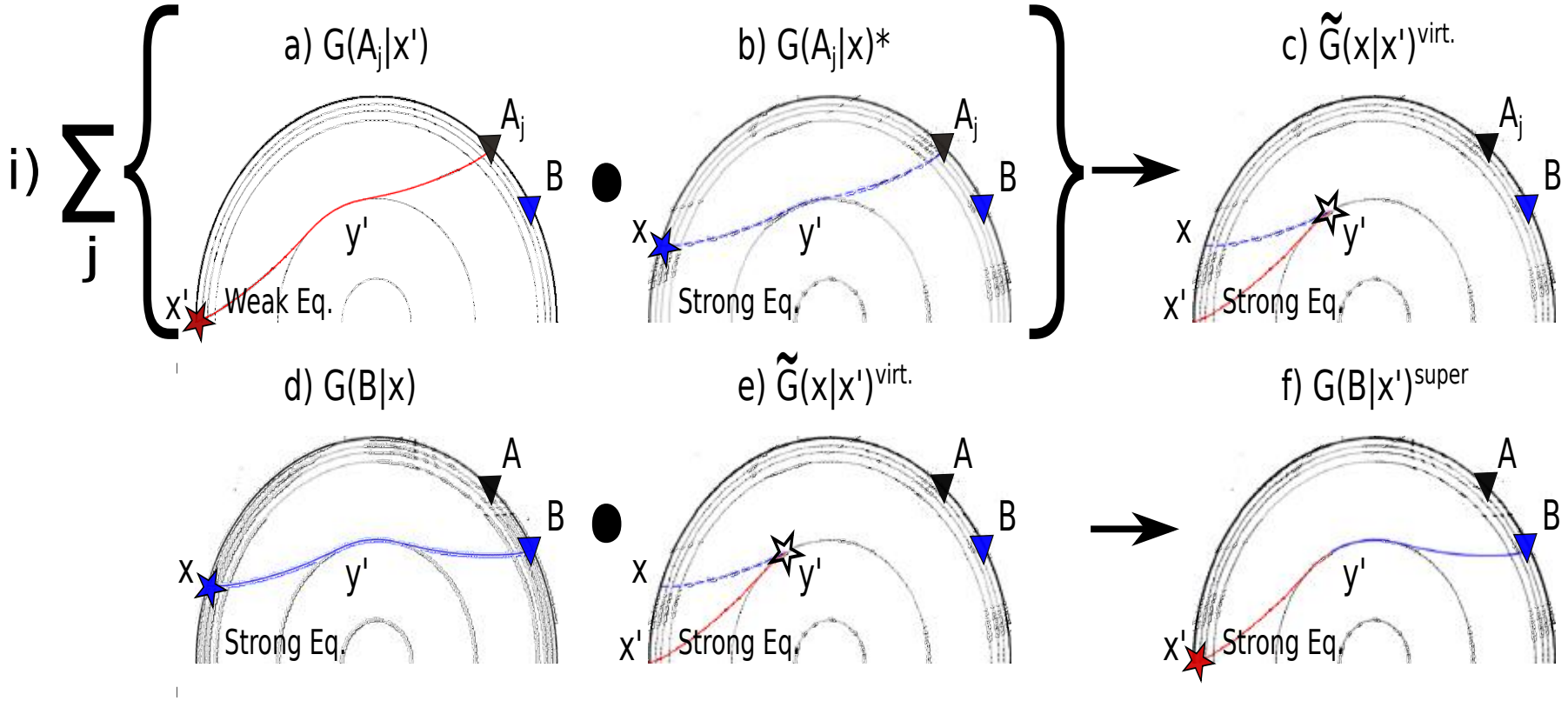
AIM: Generate virtual records of ★ event at blue stations

Convolution



AIM: Generate virtual records of ★ event at blue stations

Convolution



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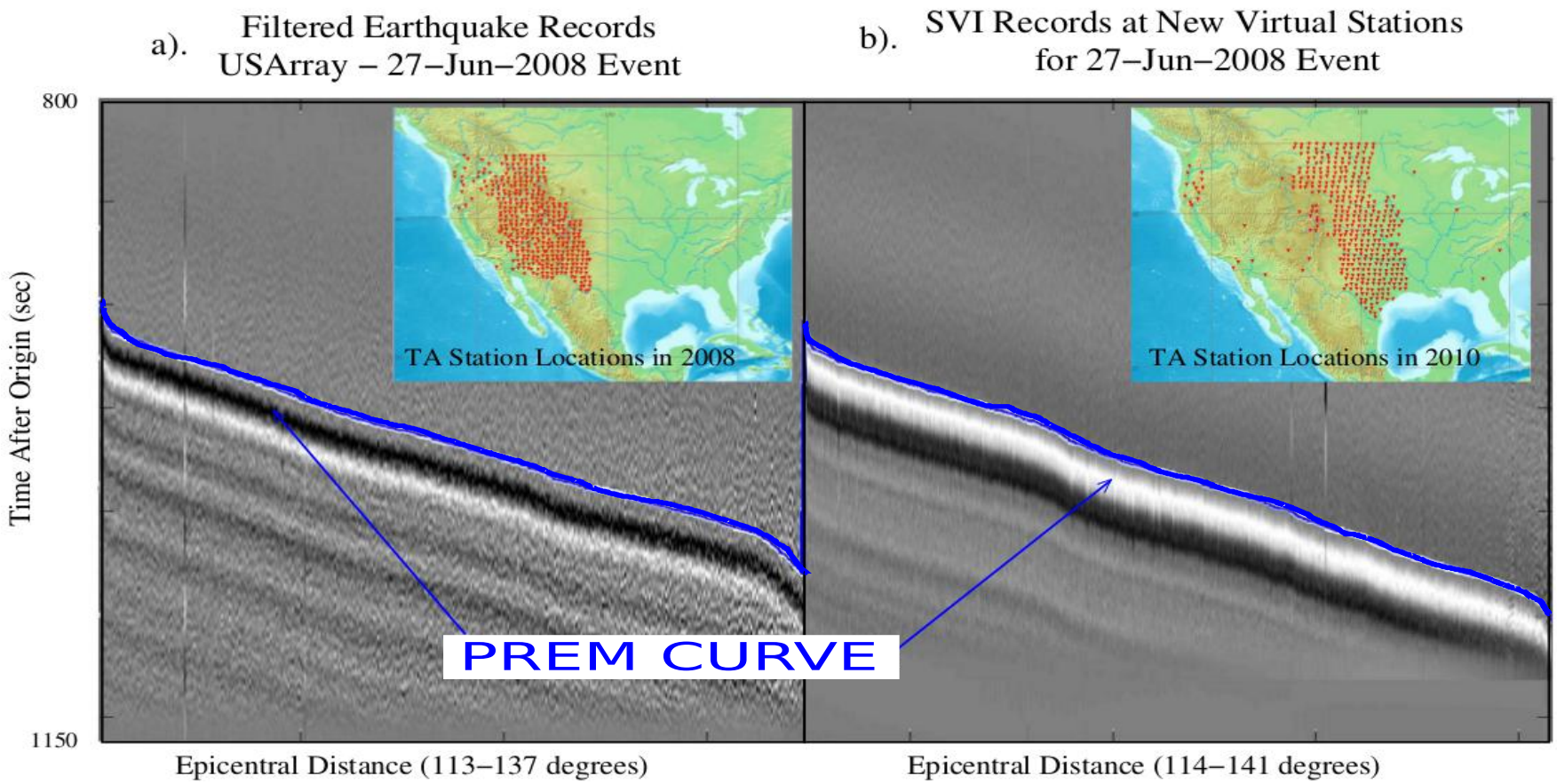
USArray Data; New Records of June 2008 Eq.



Results are only kinematically correct.

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Conclusions

- Super-virtual Interferometry: Datuming by cross-correlation and Dedatuming by convolution
- Benefits
  - Enhancing P and S diff arrivals due to weak events by $\sqrt{N_g}$
  - Generation of new virtual recordings
- Limitations
  - Spatial sources only lead to spatial
  - Kinematics OK but amplitudes
  - Source-receiver pairs only for which reflection/diffraction exists
  - Over-sampling in the source-receiver domain
  - Wave-equation based methods are computationally more expensive

Conclusions

- Super-virtual Interferometry: Datuming by cross-correlation and Dedatuming by convolution
- Benefits
  - Enhancing P and S diff arrivals due to weak events by $\sqrt{N_g}$
  - Generation of new virtual recordings
- Limitations
  - Strong source may leave imprint on result
  - Assumptions OK for 2D cases
  - Extrapolation of Green's function for dedatuming in 3D case is not trivial
  - Over-sampling in the source coordinates is necessary for dedatuming

Conclusions

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  - Enhancing P and S diff arrivals due to weak events by $\sqrt{N_g}$
  - Generation of new virtual recordings
- Limitations
  - Strong source may leave imprint on result
  - Kinematics OK, but amplitudes not
  - Source signature is more complex for diffractions and hence difficult to correct for after deconvolution
  - Very sensitive to noise in the source-receiver data, hence difficult to correct

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  - Weak events epicenter on the same great circle as that of strong event epicenter

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Future Work - Inverse Problem

Minimize: $J = \sum_{\omega}(\chi(D_{weak}F, D_{strong}) - D_{strong})^2$

F = unknown filter which improves the SNR
χ = Super-virtual function (cross-correlation & convolution)
D_{weak} = Diffracted arrivals due to weak Eq.
D_{strong} = Diffracted arrivals due to strong Eq.

Output SVI records ($D_{weak}F$) are deconvolved and more spiky !

Acknowledgements: Tarje Nissen-Meyer for providing the teleseismic synthetic data. CSIM consortium for the financial support during my internship at KAUST. IRIS for maintaining downloadable online USArray database.

Thank you ! Questions ?